Fluidum: the derivations of the generated matrices

Each section is written by the derivation script it describes, from the matrices, sources and check results that script computed:

- `deriv/derive_2p1d_viscous.wls`
- `deriv/derive_hq_2p1d.wls`
- `deriv/derive_two_way.wls`

Regenerate with `deriv/render_all.sh`.

Conventions: boost-invariant Milne coordinates; every system is written source on the left, $A_t\,\partial_\tau\phi + A_x\,\partial_x\phi + A_y\,\partial_y\phi + S = 0$; only nonzero entries are listed; $u^\tau = \sqrt{1+u_\perp^2}$ is written for the root wherever it occurs.

Contents

1 Viscous hydrodynamics: Israel–Stewart with $\delta_{\pi\pi}$ **2**

1.1 Setup . . . 2

1.2 1+1D cylindrical . . . 2

1.3 2+1D Cartesian . . . 4

1.4 Checks . . . 9

2 Charm first moment on a 2+1D background ($\nabla\alpha$ closure) **10**

2.1 Setup . . . 10

2.2 Result . . . 10

2.3 Checks . . . 12

3 Medium, charm and the two-way coupled system **13**

3.1 Setup . . . 13

3.2 Result . . . 14

3.3 1+1D cylindrical . . . 14

3.4 2+1D Cartesian . . . 20

3.5 Checks . . . 22

1 Viscous hydrodynamics: Israel–Stewart with $\delta_{\pi\pi}$

Script: `deriv/derive_2p1d_viscous.wls`

Writes: `two_d_viscous_matrix_derived` in `src/Matrix/2d_viscous.jl`; the 1+1D system is the control, compared entry by entry with `one_d_viscous_matrix` (V-G4).

1.1 Setup

Milne coordinates, boost invariant. Two charts:

$$x^\mu = (\tau, r, \phi, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, r^2, \tau^2)$$
$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2)$$

$$T^{\mu\nu} = (e + p + \Pi)\, u^\mu u^\nu + (p + \Pi)\, g^{\mu\nu} + \pi^{\mu\nu}, \qquad e = T\, p'(T) - p(T)$$
$$\nabla_\mu T^{\mu\nu} = 0$$
$$\tau_\pi\, \Delta^{\mu\nu}_{\alpha\beta}\, D\pi^{\alpha\beta} + \pi^{\mu\nu} + 2\eta\, \sigma^{\mu\nu} + \delta_{\pi\pi}\, \pi^{\mu\nu}\theta = 0$$
$$\tau_\Pi\, D\Pi + \Pi + \zeta\, \theta = 0$$

with

$$D = u^\mu \nabla_\mu, \qquad \theta = \nabla_\mu u^\mu, \qquad \Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$$
$$\sigma^{\mu\nu} = \Delta^{\mu\nu}_{\alpha\beta} \nabla^\alpha u^\beta, \qquad \Delta^{\mu\nu}_{\alpha\beta} = \tfrac{1}{2}\left(\Delta^\mu_\alpha \Delta^\nu_\beta + \Delta^\mu_\beta \Delta^\nu_\alpha\right) - \tfrac{1}{3}\Delta^{\mu\nu}\Delta_{\alpha\beta}$$

η, τ_π, ζ, τ_Π are not differentiated. Quasi-linear form, source on the left:

$$A_t\, \partial_\tau \phi + A_r\, \partial_r \phi + S = 0 \quad \text{(1+1D)}$$
$$A_t\, \partial_\tau \phi + A_x\, \partial_x \phi + A_y\, \partial_y \phi + S = 0 \quad \text{(2+1D)}$$

The $\delta_{\pi\pi}$ term is linear, so each matrix and the source split as

$$A = A^{\mathrm{IS}} + \delta_{\pi\pi}\, A^{(\delta)}, \qquad S = S^{\mathrm{IS}} + \delta_{\pi\pi}\, S^{(\delta)}$$

1.2 1+1D cylindrical

Fields, with π^{rr} and the τ row of $\pi^{\mu\nu}$ fixed by $u_\mu \pi^{\mu\nu} = 0$, $\pi^\mu{}_\mu = 0$:

$$\phi = (T, u^r, \pi^\phi{}_\phi, \pi^\eta{}_\eta, \Pi), \qquad \pi^\phi{}_\phi = r^2 \pi^{\phi\phi}, \quad \pi^\eta{}_\eta = \tau^2 \pi^{\eta\eta}$$
$$\pi^{rr} = -\left((u^\tau)^2\, (\pi^\eta{}_\eta + \pi^\phi{}_\phi)\right), \qquad u^\tau = \sqrt{(u^r)^2 + 1}$$

Rows: $\nabla_\mu T^{\mu\tau}$, $\nabla_\mu T^{\mu r}$, $E^{\phi\phi}$, $E^{\eta\eta}$, E_Π, where $E^{\mu\nu}$ and E_Π are the left-hand sides of the shear and bulk equations.

1.2.1 IS part

$$(A_t^{\mathrm{IS}})_{11} = (u^r)^2\, (p''(T) T + p'(T)) + p''(T) T$$
$$(A_t^{\mathrm{IS}})_{12} = 2u^r (p'(T) T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi)$$
$$(A_t^{\mathrm{IS}})_{13} = -(u^r)^2$$
$$(A_t^{\mathrm{IS}})_{14} = -(u^r)^2$$
$$(A_t^{\mathrm{IS}})_{15} = (u^r)^2$$

$$
\begin{aligned}
(A_t^{\mathrm{IS}})_{21} &= u^r u^\tau (p''(T)T + p'(T)) \\
(A_t^{\mathrm{IS}})_{22} &= \frac{\left(2\left(u^r\right)^2 + 1\right)\left(p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi\right)}{u^\tau} \\
(A_t^{\mathrm{IS}})_{23} &= -u^r u^\tau \\
(A_t^{\mathrm{IS}})_{24} &= -u^r u^\tau \\
(A_t^{\mathrm{IS}})_{25} &= u^r u^\tau \\
(A_t^{\mathrm{IS}})_{32} &= -\frac{2u^r \eta}{3r^2 u^\tau} \\
(A_t^{\mathrm{IS}})_{33} &= \frac{\tau_\pi u^\tau}{r^2} \\
(A_t^{\mathrm{IS}})_{42} &= -\frac{2u^r \eta}{3\tau^2 u^\tau} \\
(A_t^{\mathrm{IS}})_{44} &= \frac{\tau_\pi u^\tau}{\tau^2} \\
(A_t^{\mathrm{IS}})_{52} &= \frac{u^r \zeta}{u^\tau} \\
(A_t^{\mathrm{IS}})_{55} &= \tau_\Pi u^\tau
\end{aligned}
$$

$$
\begin{aligned}
(A_r^{\mathrm{IS}})_{11} &= u^r u^\tau (p''(T)T + p'(T)) \\
(A_r^{\mathrm{IS}})_{12} &= \frac{\left(2\left(u^r\right)^2 + 1\right)\left(p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi\right)}{u^\tau} \\
(A_r^{\mathrm{IS}})_{13} &= -u^r u^\tau \\
(A_r^{\mathrm{IS}})_{14} &= -u^r u^\tau \\
(A_r^{\mathrm{IS}})_{15} &= u^r u^\tau \\
(A_r^{\mathrm{IS}})_{21} &= p''(T)\left(u^r\right)^2 T + p'(T)\left(u^r\right)^2 + p'(T) \\
(A_r^{\mathrm{IS}})_{22} &= 2u^r (p'(T)T + \Pi - \pi^\eta{}_\eta - \pi^\phi{}_\phi) \\
(A_r^{\mathrm{IS}})_{23} &= -\left(u^r\right)^2 - 1 \\
(A_r^{\mathrm{IS}})_{24} &= -\left(u^r\right)^2 - 1 \\
(A_r^{\mathrm{IS}})_{25} &= \left(u^\tau\right)^2 \\
(A_r^{\mathrm{IS}})_{32} &= -\frac{2\eta}{3r^2} \\
(A_r^{\mathrm{IS}})_{33} &= \frac{\tau_\pi u^r}{r^2} \\
(A_r^{\mathrm{IS}})_{42} &= -\frac{2\eta}{3\tau^2} \\
(A_r^{\mathrm{IS}})_{44} &= \frac{\tau_\pi u^r}{\tau^2} \\
(A_r^{\mathrm{IS}})_{52} &= \zeta \\
(A_r^{\mathrm{IS}})_{55} &= \tau_\Pi u^r
\end{aligned}
$$

$$
\begin{aligned}
S_1^{\mathrm{IS}} &= \frac{1}{r\tau u^\tau}\Big(\left(u^\tau\right)^2 (p'(T)T + \Pi)(ru^\tau + \tau u^r) - \pi^\phi{}_\phi u^r \left(ru^r u^\tau + \tau\left(u^r\right)^2 + \tau\right) \\
&\quad - \pi^\eta{}_\eta \left(r\left(\left(u^r\right)^2 - 1\right)u^\tau + \tau\left(\left(u^r\right)^3 + u^r\right)\right)\Big) \\
S_2^{\mathrm{IS}} &= \frac{1}{r\tau u^\tau}\Big(-\pi^\eta{}_\eta \left(u^\tau\right)^2 (ru^r + \tau u^\tau) + u^r (p'(T)T + \Pi)\left(r\left(u^r\right)^2 + r + \tau u^r u^\tau\right) \\
&\quad - \pi^\phi{}_\phi \left(r\left(\left(u^r\right)^3 + u^r\right) + \tau\left(\left(u^r\right)^2 + 2\right)u^\tau\right)\Big) \\
S_3^{\mathrm{IS}} &= \frac{\pi^\phi{}_\phi}{r^2} - \frac{2\eta\left(r\left(u^r\right)^2 + r - 2\tau u^r u^\tau\right)}{3r^3 \tau u^\tau} \\
S_4^{\mathrm{IS}} &= \frac{3\pi^\eta{}_\eta \tau - \frac{2\tau u^r \eta}{r} + 4u^\tau \eta}{3\tau^3}
\end{aligned}
$$

$$S_5^{\mathrm{IS}} = \Pi + \frac{u^r \zeta}{r} + \frac{u^\tau \zeta}{\tau}$$

1.2.2 $\delta_{\pi\pi}$ part

$$(A_t^{(\delta)})_{32} = \frac{\pi^\phi{}_\phi u^r}{r^2 u^\tau}$$
$$(A_t^{(\delta)})_{42} = \frac{\pi^\eta{}_\eta u^r}{\tau^2 u^\tau}$$

$$(A_r^{(\delta)})_{32} = \frac{\pi^\phi{}_\phi}{r^2}$$
$$(A_r^{(\delta)})_{42} = \frac{\pi^\eta{}_\eta}{\tau^2}$$

$$S_3^{(\delta)} = \frac{\pi^\phi{}_\phi (r u^\tau + \tau u^r)}{r^3 \tau}$$
$$S_4^{(\delta)} = \frac{\pi^\eta{}_\eta (r u^\tau + \tau u^r)}{r \tau^3}$$

1.3 2+1D Cartesian

Fields, with π^{xx} and the τ row fixed by the same constraints:

$$\phi = (T, u^x, u^y, \pi^{yy}, \pi^\eta{}_\eta, \pi^{xy}, \Pi)$$
$$\pi^{xx} = -\frac{\pi^\eta{}_\eta \left(u^\tau\right)^2 - 2\pi^{xy} u^x u^y + \pi^{yy} \left(u^x\right)^2 + \pi^{yy}}{\left(u^y\right)^2 + 1}, \qquad u^\tau = \sqrt{\left(u^x\right)^2 + \left(u^y\right)^2 + 1}$$

Rows: $\nabla_\mu T^{\mu\tau}$, $\nabla_\mu T^{\mu x}$, $\nabla_\mu T^{\mu y}$, E^{yy}, $E^{\eta\eta}$, E^{xy}, E_Π.

1.3.1 IS part

$$(A_t^{\mathrm{IS}})_{11} = \left(u^\tau\right)^2 \left(p''(T)T + p'(T)\right) - p'(T)$$
$$(A_t^{\mathrm{IS}})_{12} = \frac{2\left(p'(T)T u^x \left(u^y\right)^2 + p'(T)T u^x + \Pi u^x \left(\left(u^y\right)^2 + 1\right) - \pi^\eta{}_\eta u^x + \pi^{xy} u^y - \pi^{yy} u^x\right)}{\left(u^y\right)^2 + 1}$$
$$(A_t^{\mathrm{IS}})_{13} = \frac{2}{\left(\left(u^y\right)^2 + 1\right)^2}\Big(\pi^{xy}\left(u^x - u^x \left(u^y\right)^2\right)$$
$$+ u^y\Big(\pi^{yy} + p'(T)T + \pi^\eta{}_\eta \left(u^x\right)^2 + \pi^{yy}\left(u^x\right)^2 + 2p'(T)T\left(u^y\right)^2 + p'(T)T\left(u^y\right)^4$$
$$+ \Pi\left(\left(u^y\right)^2 + 1\right)^2\Big)\Big)$$
$$(A_t^{\mathrm{IS}})_{14} = \frac{\left(u^y\right)^2 - \left(u^x\right)^2}{\left(u^y\right)^2 + 1}$$
$$(A_t^{\mathrm{IS}})_{15} = -\frac{\left(u^x\right)^2}{\left(u^y\right)^2 + 1}$$
$$(A_t^{\mathrm{IS}})_{16} = \frac{2u^x u^y}{\left(u^y\right)^2 + 1}$$
$$(A_t^{\mathrm{IS}})_{17} = \left(u^x\right)^2 + \left(u^y\right)^2$$
$$(A_t^{\mathrm{IS}})_{21} = u^\tau u^x (p''(T)T + p'(T))$$
$$(A_t^{\mathrm{IS}})_{22} = \frac{1}{\left(u^\tau\right)^3}\Big(\left(u^\tau\right)^4 \left(p'(T)T + \Pi\right) + \left(u^\tau\right)^2 \left(u^x\right)^2 \left(p'(T)T + \Pi\right)$$

$$+\frac{u^x\left(\pi^\eta{}_\eta\left(u^\tau\right)^2u^x-\pi^{xy}u^y\left(2\left(u^x\right)^2+\left(u^y\right)^2+1\right)+\pi^{yy}\left(\left(u^x\right)^3+u^x\right)\right)}{\left(u^y\right)^2+1}$$
$$-\frac{\left(u^\tau\right)^2\left(\pi^\eta{}_\eta\left(3\left(u^x\right)^2+\left(u^y\right)^2+1\right)-4\pi^{xy}u^xu^y+3\pi^{yy}\left(u^x\right)^2+\pi^{yy}\right)}{\left(u^y\right)^2+1}\Bigg)$$

$$(A_t^{\mathrm{IS}})_{23}=\frac{1}{\left(u^\tau\right)^3}\Bigg(\left(u^\tau\right)^2u^xu^y(p'(T)T+\Pi)+u^y\left(\frac{u^x\left(\pi^\eta{}_\eta\left(u^\tau\right)^2-2\pi^{xy}u^xu^y+\pi^{yy}\left(u^x\right)^2+\pi^{yy}\right)}{\left(u^y\right)^2+1}-\pi^{xy}u^y\right)$$
$$+\frac{\left(u^\tau\right)^2\left(2u^xu^y\left(\pi^\eta{}_\eta\left(u^x\right)^2+\pi^{yy}\left(u^x\right)^2+\pi^{yy}\right)+\pi^{xy}\left(\left(\left(u^y\right)^2+1\right)^2-2\left(u^x\right)^2\left(\left(u^y\right)^2-1\right)\right)\right)}{\left(\left(u^y\right)^2+1\right)^2}\Bigg)$$

$$(A_t^{\mathrm{IS}})_{24}=-\frac{u^x\left(\left(u^x\right)^2+1\right)}{u^\tau\left(\left(u^y\right)^2+1\right)}$$
$$(A_t^{\mathrm{IS}})_{25}=-\frac{u^\tau u^x}{\left(u^y\right)^2+1}$$
$$(A_t^{\mathrm{IS}})_{26}=\frac{u^y\left(\frac{2\left(u^x\right)^2}{\left(u^y\right)^2+1}+1\right)}{u^\tau}$$
$$(A_t^{\mathrm{IS}})_{27}=u^\tau u^x$$
$$(A_t^{\mathrm{IS}})_{31}=u^\tau u^y(p''(T)T+p'(T))$$
$$(A_t^{\mathrm{IS}})_{32}=\frac{u^xu^y\left(p'(T)T\left(u^\tau\right)^2+\Pi\left(u^\tau\right)^2-\pi^{yy}\right)+\pi^{xy}\left(\left(u^y\right)^2+1\right)}{\left(u^\tau\right)^3}$$
$$(A_t^{\mathrm{IS}})_{33}=\frac{\left(u^\tau\right)^4(p'(T)T+\Pi)+\left(u^\tau\right)^2\left(u^y\right)^2(p'(T)T+\Pi)-u^y(\pi^{xy}u^x+\pi^{yy}u^y)+\pi^{yy}\left(u^\tau\right)^2}{\left(u^\tau\right)^3}$$
$$(A_t^{\mathrm{IS}})_{34}=\frac{u^y}{u^\tau}$$
$$(A_t^{\mathrm{IS}})_{36}=\frac{u^x}{u^\tau}$$
$$(A_t^{\mathrm{IS}})_{37}=u^\tau u^y$$
$$(A_t^{\mathrm{IS}})_{42}=-\frac{2\left(3\pi^{xy}\tau_\pi u^y\left(\left(u^y\right)^2+1\right)+u^x\left(\left(u^y\right)^2(\eta-3\pi^{yy}\tau_\pi)+\eta\right)\right)}{3u^\tau}$$
$$(A_t^{\mathrm{IS}})_{43}=\frac{2u^y\left(3\pi^{xy}\tau_\pi u^xu^y-3\pi^{yy}\tau_\pi\left(\left(u^x\right)^2+1\right)+\eta\left(3\left(u^x\right)^2+2\left(u^y\right)^2+2\right)\right)}{3u^\tau}$$
$$(A_t^{\mathrm{IS}})_{44}=\tau_\pi u^\tau$$
$$(A_t^{\mathrm{IS}})_{52}=-\frac{2u^x\eta}{3\tau^2u^\tau}$$
$$(A_t^{\mathrm{IS}})_{53}=-\frac{2u^y\eta}{3\tau^2u^\tau}$$
$$(A_t^{\mathrm{IS}})_{55}=\frac{\tau_\pi u^\tau}{\tau^2}$$
$$(A_t^{\mathrm{IS}})_{62}=\frac{1}{3u^\tau}\Big(-3\pi^{xy}\tau_\pi u^x\left(2\left(u^y\right)^2+1\right)$$
$$+u^y\left(3\pi^\eta{}_\eta\tau_\pi\left(u^\tau\right)^2+3\pi^{yy}\left(2\tau_\pi\left(u^x\right)^2+\tau_\pi\right)+\eta\left(\left(u^x\right)^2+3\left(u^y\right)^2+3\right)\right)\Big)$$
$$(A_t^{\mathrm{IS}})_{63}=\frac{1}{3u^\tau\left(\left(u^y\right)^2+1\right)}\Big(-3\pi^{yy}\tau_\pi u^x\left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right)+3\pi^{xy}\tau_\pi u^y\left(\left(2\left(u^x\right)^2-1\right)\left(u^y\right)^2-1\right)$$
$$+u^x\left(\left(\left(u^y\right)^2+1\right)\eta\left(3\left(u^x\right)^2+\left(u^y\right)^2+3\right)-3\pi^\eta{}_\eta\tau_\pi\left(u^\tau\right)^2\left(u^y\right)^2\right)\Big)$$
$$(A_t^{\mathrm{IS}})_{66}=\tau_\pi u^\tau$$
$$(A_t^{\mathrm{IS}})_{72}=\frac{u^x\zeta}{u^\tau}$$

$$(A_t^{\mathrm{IS}})_{73} = \frac{u^y \zeta}{u^\tau}$$
$$(A_t^{\mathrm{IS}})_{77} = \tau_\Pi u^\tau$$

$$(A_x^{\mathrm{IS}})_{11} = u^\tau u^x (p''(T)T + p'(T))$$
$$(A_x^{\mathrm{IS}})_{12} = \frac{1}{(u^\tau)^3}\Big((u^\tau)^4 (p'(T)T + \Pi) + (u^\tau)^2 (u^x)^2 (p'(T)T + \Pi) + \frac{u^x \left(\pi^\eta{}_\eta (u^\tau)^2 u^x - \pi^{xy} u^y \left(2(u^x)^2 + (u^y)^2 + 1\right) + \pi^{yy}\left((u^x)^3 + u^x\right)\right)}{(u^y)^2 + 1} - \frac{(u^\tau)^2 \left(\pi^\eta{}_\eta \left(3(u^x)^2 + (u^y)^2 + 1\right) - 4\pi^{xy} u^x u^y + 3\pi^{yy}(u^x)^2 + \pi^{yy}\right)}{(u^y)^2 + 1}\Big)$$
$$(A_x^{\mathrm{IS}})_{13} = \frac{1}{(u^\tau)^3}\Big((u^\tau)^2 u^x u^y (p'(T)T + \Pi) + u^y \left(\frac{u^x \left(\pi^\eta{}_\eta (u^\tau)^2 - 2\pi^{xy} u^x u^y + \pi^{yy}(u^x)^2 + \pi^{yy}\right)}{(u^y)^2 + 1} - \pi^{xy} u^y\right) + \frac{(u^\tau)^2 \left(2u^x u^y \left(\pi^\eta{}_\eta (u^x)^2 + \pi^{yy}(u^x)^2 + \pi^{yy}\right) + \pi^{xy}\left(\left((u^y)^2 + 1\right)^2 - 2(u^x)^2\left((u^y)^2 - 1\right)\right)\right)}{\left((u^y)^2 + 1\right)^2}\Big)$$
$$(A_x^{\mathrm{IS}})_{14} = -\frac{u^x \left((u^x)^2 + 1\right)}{u^\tau \left((u^y)^2 + 1\right)}$$
$$(A_x^{\mathrm{IS}})_{15} = -\frac{u^\tau u^x}{(u^y)^2 + 1}$$
$$(A_x^{\mathrm{IS}})_{16} = \frac{u^y \left(\frac{2(u^x)^2}{(u^y)^2+1} + 1\right)}{u^\tau}$$
$$(A_x^{\mathrm{IS}})_{17} = u^\tau u^x$$
$$(A_x^{\mathrm{IS}})_{21} = p''(T)T (u^x)^2 + p'(T)(u^x)^2 + p'(T)$$
$$(A_x^{\mathrm{IS}})_{22} = 2u^x (p'(T)T + \Pi) - \frac{2(\pi^\eta{}_\eta u^x - \pi^{xy} u^y + \pi^{yy} u^x)}{(u^y)^2 + 1}$$
$$(A_x^{\mathrm{IS}})_{23} = \frac{2\left(u^y \left(\pi^\eta{}_\eta (u^x)^2 + \pi^{yy}(u^x)^2 + \pi^{yy}\right) + \pi^{xy}\left(u^x - u^x (u^y)^2\right)\right)}{\left((u^y)^2 + 1\right)^2}$$
$$(A_x^{\mathrm{IS}})_{24} = -\frac{(u^x)^2 + 1}{(u^y)^2 + 1}$$
$$(A_x^{\mathrm{IS}})_{25} = -\frac{(u^\tau)^2}{(u^y)^2 + 1}$$
$$(A_x^{\mathrm{IS}})_{26} = \frac{2u^x u^y}{(u^y)^2 + 1}$$
$$(A_x^{\mathrm{IS}})_{27} = (u^x)^2 + 1$$
$$(A_x^{\mathrm{IS}})_{31} = u^x u^y (p''(T)T + p'(T))$$
$$(A_x^{\mathrm{IS}})_{32} = u^y (p'(T)T + \Pi)$$
$$(A_x^{\mathrm{IS}})_{33} = u^x (p'(T)T + \Pi)$$
$$(A_x^{\mathrm{IS}})_{36} = 1$$
$$(A_x^{\mathrm{IS}})_{37} = u^x u^y$$
$$(A_x^{\mathrm{IS}})_{42} = -\frac{2\left(3\pi^{xy}\tau_\pi u^x u^y \left((u^y)^2 + 1\right) - 3\pi^{yy}\tau_\pi (u^x)^2 (u^y)^2 + (u^\tau)^2 \left((u^y)^2 + 1\right)\eta\right)}{3(u^\tau)^2}$$
$$(A_x^{\mathrm{IS}})_{43} = \frac{2u^x u^y \left(\pi^{xy}\tau_\pi u^x u^y - \pi^{yy}\tau_\pi \left((u^x)^2 + 1\right) + (u^\tau)^2 \eta\right)}{(u^\tau)^2}$$

$$(A_x^{\mathrm{IS}})_{44} = \tau_\pi u^x$$
$$(A_x^{\mathrm{IS}})_{52} = -\frac{2\eta}{3\tau^2}$$
$$(A_x^{\mathrm{IS}})_{55} = \frac{\tau_\pi u^x}{\tau^2}$$
$$(A_x^{\mathrm{IS}})_{62} = \frac{u^x\left(u^y\left(3\pi^\eta{}_\eta\tau_\pi\left(u^\tau\right)^2+3\pi^{yy}\left(2\tau_\pi\left(u^x\right)^2+\tau_\pi\right)+\left(u^\tau\right)^2\eta\right)-3\pi^{xy}\tau_\pi u^x\left(2\left(u^y\right)^2+1\right)\right)}{3\left(u^\tau\right)^2}$$
$$(A_x^{\mathrm{IS}})_{63} = \frac{1}{\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}\Big(-\pi^{yy}\tau_\pi\left(u^x\right)^2\left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right)+\pi^{xy}\tau_\pi u^x u^y\left(\left(2\left(u^x\right)^2-1\right)\left(u^y\right)^2-1\right)$$
$$-\left(u^\tau\right)^2\left(\pi^\eta{}_\eta\tau_\pi\left(u^x\right)^2\left(u^y\right)^2-\left(\left(u^x\right)^2+1\right)\left(\left(u^y\right)^2+1\right)\eta\right)\Big)$$
$$(A_x^{\mathrm{IS}})_{66} = \tau_\pi u^x$$
$$(A_x^{\mathrm{IS}})_{72} = \zeta$$
$$(A_x^{\mathrm{IS}})_{77} = \tau_\Pi u^x$$

$$(A_y^{\mathrm{IS}})_{11} = u^\tau u^y(p''(T)T+p'(T))$$
$$(A_y^{\mathrm{IS}})_{12} = \frac{u^x u^y\left(p'(T)T\left(u^\tau\right)^2+\Pi\left(u^\tau\right)^2-\pi^{yy}\right)+\pi^{xy}\left(\left(u^y\right)^2+1\right)}{\left(u^\tau\right)^3}$$
$$(A_y^{\mathrm{IS}})_{13} = \frac{\left(u^\tau\right)^4\left(p'(T)T+\Pi\right)+\left(u^\tau\right)^2\left(u^y\right)^2\left(p'(T)T+\Pi\right)-u^y(\pi^{xy}u^x+\pi^{yy}u^y)+\pi^{yy}\left(u^\tau\right)^2}{\left(u^\tau\right)^3}$$
$$(A_y^{\mathrm{IS}})_{14} = \frac{u^y}{u^\tau}$$
$$(A_y^{\mathrm{IS}})_{16} = \frac{u^x}{u^\tau}$$
$$(A_y^{\mathrm{IS}})_{17} = u^\tau u^y$$
$$(A_y^{\mathrm{IS}})_{21} = u^x u^y(p''(T)T+p'(T))$$
$$(A_y^{\mathrm{IS}})_{22} = u^y(p'(T)T+\Pi)$$
$$(A_y^{\mathrm{IS}})_{23} = u^x(p'(T)T+\Pi)$$
$$(A_y^{\mathrm{IS}})_{26} = 1$$
$$(A_y^{\mathrm{IS}})_{27} = u^x u^y$$
$$(A_y^{\mathrm{IS}})_{31} = p''(T)T\left(u^y\right)^2+p'(T)\left(u^y\right)^2+p'(T)$$
$$(A_y^{\mathrm{IS}})_{33} = 2u^y(p'(T)T+\Pi)$$
$$(A_y^{\mathrm{IS}})_{34} = 1$$
$$(A_y^{\mathrm{IS}})_{37} = \left(u^y\right)^2+1$$
$$(A_y^{\mathrm{IS}})_{42} = -\frac{2\tau_\pi\left(u^y\right)^2\left(\pi^{xy}\left(u^y\right)^2+\pi^{xy}-\pi^{yy}u^x u^y\right)}{\left(u^\tau\right)^2}$$
$$(A_y^{\mathrm{IS}})_{43} = \frac{6\pi^{xy}\tau_\pi u^x\left(u^y\right)^3-6\pi^{yy}\tau_\pi\left(\left(u^x\right)^2+1\right)\left(u^y\right)^2+4\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)\eta}{3\left(u^\tau\right)^2}$$
$$(A_y^{\mathrm{IS}})_{44} = \tau_\pi u^y$$
$$(A_y^{\mathrm{IS}})_{53} = -\frac{2\eta}{3\tau^2}$$
$$(A_y^{\mathrm{IS}})_{55} = \frac{\tau_\pi u^y}{\tau^2}$$
$$(A_y^{\mathrm{IS}})_{62} = \frac{1}{\left(u^\tau\right)^2}\Big(-\pi^{xy}\tau_\pi u^x u^y\left(2\left(u^y\right)^2+1\right)+\eta+\left(u^y\right)^4\left(\pi^\eta{}_\eta\tau_\pi+\eta\right)+\left(u^y\right)^2\left(\pi^\eta{}_\eta\tau_\pi+\pi^{yy}\tau_\pi+2\eta\right)$$
$$+\left(u^x\right)^2\left(\left(u^y\right)^2\left(\pi^\eta{}_\eta\tau_\pi+2\pi^{yy}\tau_\pi+\eta\right)+\eta\right)\Big)$$

$$
\begin{aligned}
(A_y^{\mathrm{IS}})_{63} &= -\frac{1}{3\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)}u^y\Big(3\pi^{yy}\tau_\pi u^x\left(\left(u^x\right)^2+1\right)\left(2\left(u^y\right)^2+1\right)+3\pi^{xy}\tau_\pi u^y\left(\left(1-2\left(u^x\right)^2\right)\left(u^y\right)^2+1\right)\\
&\quad+\left(u^\tau\right)^2u^x\left(3\pi^\eta{}_\eta\tau_\pi\left(u^y\right)^2-\left(\left(u^y\right)^2+1\right)\eta\right)\Big)\\
(A_y^{\mathrm{IS}})_{66} &= \tau_\pi u^y\\
(A_y^{\mathrm{IS}})_{73} &= \zeta\\
(A_y^{\mathrm{IS}})_{77} &= \tau_\Pi u^y
\end{aligned}
$$

$$
\begin{aligned}
S_1^{\mathrm{IS}} &= \frac{1}{\tau\left(\left(u^y\right)^2+1\right)}\Big(p'(T)T-\pi^{yy}\left(u^x\right)^2+p'(T)T\left(u^x\right)^2+2\pi^{xy}u^xu^y+\pi^{yy}\left(u^y\right)^2+2p'(T)T\left(u^y\right)^2\\
&\quad+p'(T)T\left(u^x\right)^2\left(u^y\right)^2+p'(T)T\left(u^y\right)^4+\Pi\left(u^\tau\right)^2\left(\left(u^y\right)^2+1\right)+\pi^\eta{}_\eta\left(-\left(u^x\right)^2+\left(u^y\right)^2+1\right)\Big)\\
S_2^{\mathrm{IS}} &= \frac{1}{\tau u^\tau\left(\left(u^y\right)^2+1\right)}\Big(-\pi^{yy}u^x+p'(T)Tu^x-\pi^\eta{}_\eta\left(u^\tau\right)^2u^x-\pi^{yy}\left(u^x\right)^3+p'(T)T\left(u^x\right)^3+\pi^{xy}u^y+2\pi^{xy}\left(u^x\right)^2u^y\\
&\quad+2p'(T)Tu^x\left(u^y\right)^2+p'(T)T\left(u^x\right)^3\left(u^y\right)^2+\pi^{xy}\left(u^y\right)^3+p'(T)Tu^x\left(u^y\right)^4\\
&\quad+\Pi\left(u^\tau\right)^2u^x\left(\left(u^y\right)^2+1\right)\Big)\\
S_3^{\mathrm{IS}} &= \frac{u^y\left(p'(T)T\left(u^\tau\right)^2+\Pi\left(u^\tau\right)^2+\pi^{yy}\right)+\pi^{xy}u^x}{\tau u^\tau}\\
S_4^{\mathrm{IS}} &= \pi^{yy}-\frac{2u^\tau\left(\left(u^y\right)^2+1\right)\eta}{3\tau}\\
S_5^{\mathrm{IS}} &= \frac{3\pi^\eta{}_\eta\tau+4u^\tau\eta}{3\tau^3}\\
S_6^{\mathrm{IS}} &= \pi^{xy}-\frac{2u^\tau u^xu^y\eta}{3\tau}\\
S_7^{\mathrm{IS}} &= \Pi+\frac{u^\tau\zeta}{\tau}
\end{aligned}
$$

1.3.2 $\delta_{\pi\pi}$ part

$$
\begin{aligned}
(A_t^{(\delta)})_{42} &= \frac{\pi^{yy}u^x}{u^\tau}\\
(A_t^{(\delta)})_{43} &= \frac{\pi^{yy}u^y}{u^\tau}\\
(A_t^{(\delta)})_{52} &= \frac{\pi^\eta{}_\eta u^x}{\tau^2u^\tau}\\
(A_t^{(\delta)})_{53} &= \frac{\pi^\eta{}_\eta u^y}{\tau^2u^\tau}\\
(A_t^{(\delta)})_{62} &= \frac{\pi^{xy}u^x}{u^\tau}\\
(A_t^{(\delta)})_{63} &= \frac{\pi^{xy}u^y}{u^\tau}
\end{aligned}
$$

$$
\begin{aligned}
(A_x^{(\delta)})_{42} &= \pi^{yy}\\
(A_x^{(\delta)})_{52} &= \frac{\pi^\eta{}_\eta}{\tau^2}\\
(A_x^{(\delta)})_{62} &= \pi^{xy}
\end{aligned}
$$

$$
\begin{aligned}
(A_y^{(\delta)})_{43} &= \pi^{yy}\\
(A_y^{(\delta)})_{53} &= \frac{\pi^\eta{}_\eta}{\tau^2}\\
(A_y^{(\delta)})_{63} &= \pi^{xy}
\end{aligned}
$$

$$S_4^{(\delta)} = \frac{\pi^{yy} u^\tau}{\tau}$$
$$S_5^{(\delta)} = \frac{\pi^{\eta}{}_{\eta} u^\tau}{\tau^3}$$
$$S_6^{(\delta)} = \frac{\pi^{xy} u^\tau}{\tau}$$

1.4 Checks

id	what it tests	
V-G0cyl	cylindrical Milne: Nabla g == 0	PASS
V-G0car	Cartesian Milne: Nabla g == 0	PASS
V-G1cyl	cylindrical: Nabla.V = X' + X/tau	PASS
V-G1car	Cartesian: Nabla.V = X' + X/tau	PASS
V-G2cyl-uu	u.u == -1	PASS
V-G2cyl-un	u_mu pi^{mu nu} == 0	PASS
V-G2cyl-tr	pi^mu_mu == 0	PASS
V-G3cyl	cylindrical system is quasi-linear	PASS
V-G6cyl	cylindrical: (IS + dpp pi theta in eq. 2) == IS + dpp * (the exported dpp part), equation by equation	PASS
V-G4	1-D: derived == shipped one_d_viscous_matrix, A_t, A_r, src entry by entry, read from the .jl, SYMBOLIC dpp [CONTROL]	PASS
V-G4-0	1-D at dpp = 0 (plain IS): plain IS == the shipped IS part	PASS
V-G4-1	1-D at dpp = 4/3 tauS (default)	PASS
V-G4m	1-D at dpp = 4/3 tauS == the medium block of the 7-field :MIS one_d_viscous_matrix_fugacity, raw, read from the .jl	PASS
V-G4m-ctl	control: at dpp = 0 it is NOT the :MIS medium block	PASS
V-G2car-uu	u.u == -1	PASS
V-G2car-un	u_mu pi^{mu nu} == 0	PASS
V-G2car-tr	pi^mu_mu == 0	PASS
V-G3car	Cartesian system is quasi-linear	PASS
V-G6car	Cartesian: (IS + dpp pi theta in eq. 2) == IS + dpp * (the exported dpp part), equation by equation	PASS
V-G5-e	energy row on Bjorken == d_tau e + (e+p+Pi+pi^eta_eta)/tau	PASS
V-G5-pi	pi^eta_eta row on Bjorken == tauS pi' + pi + 4 eta/(3 tau) + dpp pi/tau (theta = 1/tau)	PASS
V-G5-Pi	bulk row on Bjorken == tauB Pi' + Pi + zeta/tau	PASS
V-G5-mom	the x, y and xy rows vanish identically on Bjorken flow	PASS

23 passed, 0 failed (23 checks evaluated).

Not checked in this script: the slab limit (2+1D at $u^y = 0$, $\partial_y = 0$ against 1+1D at $r \to \infty$); it is a numerical comparison of the two generated routines.

Re-run with `wolframscript -file deriv/derive_2p1d_viscous.wls`.

2 Charm first moment on a 2+1D background ($\nabla\alpha$ closure)

Script: `deriv/derive_hq_2p1d.wls`

Writes: `two_d_HQ_BG_matrix` and `two_d_HQ_live_rows` in `src/Matrix/2d_viscous_HQ_BG.jl`

2.1 Setup

Cartesian Milne coordinates, boost invariant, on a prescribed background T, u^x, u^y:

$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2), \qquad u^\mu = (u^\tau, u^x, u^y, 0), \quad u^\tau = \sqrt{(u^x)^2 + (u^y)^2 + 1}$$

Charm is a passive tracer:

$$N^\mu = n\, u^\mu + \nu^\mu, \qquad \nabla_\mu N^\mu = 0$$
$$\tau_n\, \Delta^\mu{}_\beta\, D\nu^\beta + \nu^\mu + \kappa\, \Delta^{\mu\beta}\partial_\beta\alpha = 0$$
$$n = n(T, \alpha), \qquad \kappa = D_s\, \partial_\alpha n, \qquad D = u^\mu \nabla_\mu, \qquad \Delta^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$$

Navier–Stokes limit: Fick's law $\nu^\mu = -\kappa\, \nabla_\perp^\mu \alpha$. Fields, with ν^τ from $u_\mu \nu^\mu = 0$:

$$\phi = (\alpha, \nu^x, \nu^y), \qquad \nu^\tau = \frac{\nu^x u^x + \nu^y u^y}{u^\tau}$$

Rows: $\nabla_\mu N^\mu$ and the x, y components of the current equation, in

$$A_t\, \partial_\tau \phi + A_x\, \partial_x \phi + A_y\, \partial_y \phi + S = 0$$

2.2 Result

$$\begin{aligned}
(A_t)_{11} &= (\partial_\alpha n) u^\tau \\
(A_t)_{12} &= \frac{u^x}{u^\tau} \\
(A_t)_{13} &= \frac{u^y}{u^\tau} \\
(A_t)_{21} &= \kappa u^\tau u^x \\
(A_t)_{22} &= \tau_n u^\tau \\
(A_t)_{31} &= \kappa u^\tau u^y \\
(A_t)_{33} &= \tau_n u^\tau
\end{aligned}$$

$$\begin{aligned}
(A_x)_{11} &= (\partial_\alpha n) u^x \\
(A_x)_{12} &= 1 \\
(A_x)_{21} &= \kappa \left((u^x)^2 + 1 \right) \\
(A_x)_{22} &= \tau_n u^x \\
(A_x)_{31} &= \kappa u^x u^y \\
(A_x)_{33} &= \tau_n u^x
\end{aligned}$$

$$\begin{aligned}
(A_y)_{11} &= (\partial_\alpha n) u^y \\
(A_y)_{13} &= 1 \\
(A_y)_{21} &= \kappa u^x u^y \\
(A_y)_{22} &= \tau_n u^y \\
(A_y)_{31} &= \kappa \left((u^y)^2 + 1 \right) \\
(A_y)_{33} &= \tau_n u^y
\end{aligned}$$

$$
\begin{aligned}
S_1 = \frac{1}{\tau\left(u^\tau\right)^3}\Big(&(\partial_\tau u^x)\nu^x\tau\left(u^\tau\right)^2 + (\partial_\tau u^y)\nu^y\tau\left(u^\tau\right)^2 + (\partial_x u^x)n\tau\left(u^\tau\right)^3 + (\partial_y u^y)n\tau\left(u^\tau\right)^3 + n\left(u^\tau\right)^4 \\
&+ (\partial_\tau T)(\partial_T n)\tau\left(u^\tau\right)^4 + \nu^x\left(u^\tau\right)^2 u^x + (\partial_\tau u^x)n\tau\left(u^\tau\right)^2 u^x + (\partial_x T)(\partial_T n)\tau\left(u^\tau\right)^3 u^x \\
&- (\partial_\tau u^x)\nu^x\tau\left(u^x\right)^2 + \nu^y\left(u^\tau\right)^2 u^y + (\partial_\tau u^y)n\tau\left(u^\tau\right)^2 u^y + (\partial_y T)(\partial_T n)\tau\left(u^\tau\right)^3 u^y \\
&- (\partial_\tau u^y)\nu^x\tau u^x u^y - (\partial_\tau u^x)\nu^y\tau u^x u^y - (\partial_\tau u^y)\nu^y\tau\left(u^y\right)^2\Big) \\
S_2 = \frac{1}{\left(u^\tau\right)^2}\Big(&-\nu^y\tau_n u^x\Big((\partial_\tau u^y)u^\tau + (\partial_x u^y)\left(u^x\right)^3 + (\partial_y u^y)u^y + \left(u^x\right)^2((\partial_\tau u^y)u^\tau - (\partial_x u^x)u^y + (\partial_y u^y)u^y) \\
&+ u^x((\partial_x u^y) - u^y((\partial_\tau u^x)u^\tau + (\partial_y u^x)u^y))\Big) \\
&+ \nu^x\Big(1 + (\partial_x u^y)\tau_n\left(u^x\right)^3 u^y + \left(u^y\right)^2 - \tau_n u^x\left(\left(u^y\right)^2 + 1\right)((\partial_\tau u^x)u^\tau + (\partial_y u^x)u^y) \\
&+ \left(u^x\right)^2\left((\partial_\tau u^y)\tau_n u^\tau u^y - (\partial_x u^x)\tau_n\left(\left(u^y\right)^2 + 1\right) + (\partial_y u^y)\tau_n\left(u^y\right)^2 + 1\right)\Big)\Big) \\
S_3 = \frac{1}{\left(u^\tau\right)^2}\Big(&-\nu^x\tau_n u^y\Big((\partial_\tau u^x)u^\tau + (\partial_x u^x)u^x + u^y((\partial_y u^x) - u^x((\partial_\tau u^y)u^\tau + (\partial_x u^y)u^x)) \\
&+ \left(u^y\right)^2((\partial_\tau u^x)u^\tau + (\partial_x u^x)u^x - (\partial_y u^y)u^x) + (\partial_y u^x)\left(u^y\right)^3\Big) \\
&+ \nu^y\Big(1 - (\partial_\tau u^y)\tau_n u^\tau u^y - (\partial_x u^y)\tau_n\left(u^x\right)^3 u^y + \left(u^y\right)^2(1 - (\partial_y u^y)\tau_n) \\
&+ \left(u^x\right)^2\left(-(\partial_\tau u^y)\tau_n u^\tau u^y + \tau_n\left(u^y\right)^2((\partial_x u^x) - (\partial_y u^y)) + 1\right) \\
&+ \tau_n u^x u^y(u^y((\partial_\tau u^x)u^\tau + (\partial_y u^x)u^y) - (\partial_x u^y))\Big)\Big)
\end{aligned}
$$

2.2.1 Live background

The same three rows with the medium promoted to fields, $\phi_L = (T, u^x, u^y, \alpha, \nu^x, \nu^y)$. Columns 4–6 are the matrices above (H-G6a); columns 1–3 and the remaining source:

$$
\begin{aligned}
(A_t^L)_{11} &= (\partial_T n)u^\tau \\
(A_t^L)_{12} &= \frac{n\left(u^\tau\right)^2 u^x + \nu^x\left(\left(u^y\right)^2 + 1\right) - \nu^y u^x u^y}{\left(u^\tau\right)^3} \\
(A_t^L)_{13} &= \frac{n\left(u^\tau\right)^2 u^y - \nu^x u^x u^y + \nu^y\left(\left(u^x\right)^2 + 1\right)}{\left(u^\tau\right)^3} \\
(A_t^L)_{22} &= \frac{\tau_n u^x\left(\nu^y u^x u^y - \nu^x\left(\left(u^y\right)^2 + 1\right)\right)}{u^\tau} \\
(A_t^L)_{23} &= -\frac{\tau_n u^x\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right)}{u^\tau} \\
(A_t^L)_{32} &= -\frac{\tau_n u^y\left(\nu^x\left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right)}{u^\tau} \\
(A_t^L)_{33} &= \frac{\tau_n u^y\left(\nu^x u^x u^y - \nu^y\left(\left(u^x\right)^2 + 1\right)\right)}{u^\tau}
\end{aligned}
$$

$$
\begin{aligned}
(A_x^L)_{11} &= (\partial_T n)u^x \\
(A_x^L)_{12} &= n \\
(A_x^L)_{22} &= \frac{\tau_n\left(u^x\right)^2\left(\nu^y u^x u^y - \nu^x\left(\left(u^y\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2} \\
(A_x^L)_{23} &= -\frac{\tau_n\left(u^x\right)^2\left(-\nu^x u^x u^y + \nu^y\left(u^x\right)^2 + \nu^y\right)}{\left(u^\tau\right)^2}
\end{aligned}
$$

$$(A_x^L)_{32} = \frac{\tau_n u^x u^y \left(\nu^y u^x u^y - \nu^x \left(\left(u^y\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$
$$(A_x^L)_{33} = \frac{\tau_n u^x u^y \left(\nu^x u^x u^y - \nu^y \left(\left(u^x\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$

$$(A_y^L)_{11} = (\partial_T n) u^y$$
$$(A_y^L)_{13} = n$$
$$(A_y^L)_{22} = \frac{\tau_n u^x u^y \left(\nu^y u^x u^y - \nu^x \left(\left(u^y\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$
$$(A_y^L)_{23} = -\frac{\tau_n u^x u^y \left(-\nu^x u^x u^y + \nu^y \left(u^x\right)^2 + \nu^y\right)}{\left(u^\tau\right)^2}$$
$$(A_y^L)_{32} = -\frac{\tau_n \left(u^y\right)^2 \left(\nu^x \left(u^y\right)^2 + \nu^x - \nu^y u^x u^y\right)}{\left(u^\tau\right)^2}$$
$$(A_y^L)_{33} = \frac{\tau_n \left(u^y\right)^2 \left(\nu^x u^x u^y - \nu^y \left(\left(u^x\right)^2 + 1\right)\right)}{\left(u^\tau\right)^2}$$

$$S_1^L = \frac{n \left(u^\tau\right)^2 + \nu^x u^x + \nu^y u^y}{\tau u^\tau}$$
$$S_2^L = \nu^x$$
$$S_3^L = \nu^y$$

2.3 Checks

id	what it tests	
`H-G0cyl`	cylindrical Milne: Nabla g == 0	PASS
`H-G0car`	Cartesian Milne: Nabla g == 0	PASS
`H-G1cyl`	cylindrical: Nabla.V = X' + X/tau for V=(X(tau),0,0,0)	PASS
`H-G1car`	Cartesian: Nabla.V = X' + X/tau	PASS
`H-G2cyl-uu`	u.u == -1	PASS
`H-G2cyl-un`	u.nu == 0	PASS
`H-G3cyl`	cylindrical system is quasi-linear	PASS
`H-G4-At`	cylindrical A_t * u^tau^3 r tau == Fluidum's shipped A_t	PASS
`H-G4-Ax`	cylindrical A_x * u^tau^3 r tau == Fluidum's shipped A_x	PASS
`H-G4-src`	cylindrical src * u^tau^3 r tau == Fluidum's shipped source	PASS
`H-G4src`	cylindrical A_t, A_x, src * u^tau^3 r tau == 1d_viscous_HQ_ cilindrical_BG.jl READ FROM THE FILE	PASS
`H-G2car-uu`	u.u == -1	PASS
`H-G2car-un`	u.nu == 0	PASS
`H-G3car`	Cartesian system is quasi-linear	PASS
`H-G5`	slab limit (u^y -> 0, d_y -> 0) decouples nu^y from (alpha, nu^x)	PASS
`H-G6a`	the (alpha, nu^x, nu^y) columns of the 10-field rows == the background-mode matrices	PASS
`H-G6b`	src_live + (medium columns).(background derivatives) == src_ background: same system, re-sorted	PASS

17 passed, 0 failed (17 checks evaluated).

Re-run with `wolframscript -file deriv/derive_hq_2p1d.wls`.

3 Medium, charm and the two-way coupled system

Script: `deriv/derive_two_way.wls`, with `derive_two_way_gates.wls` and `derive_two_way_export.wls`
Writes: the charm and two-way matrices of `src/Matrix` (one function per system, listed below)

3.1 Setup

Three charts, boost invariant: 0+1D Bjorken (fields of τ only), 1+1D cylindrical, 2+1D Cartesian Milne:

$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2), \quad u^\mu = (1, 0, 0, 0)$$
$$x^\mu = (\tau, r, \phi, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, r^2, \tau^2)$$
$$x^\mu = (\tau, x, y, \eta), \qquad g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, \tau^2)$$

Medium, with the EOS $P(T)$ kept symbolic:

$$T_m^{\mu\nu} = (e + P + \Pi)\, u^\mu u^\nu + (P + \Pi)\, g^{\mu\nu} + \pi^{\mu\nu}, \qquad e = TP'(T) - P(T)$$
$$\tau_\pi\, \Delta^{\mu\nu}_{\alpha\beta} D\pi^{\alpha\beta} + \pi^{\mu\nu} + 2\eta\, \sigma^{\mu\nu} + \delta_{\pi\pi}\, \pi^{\mu\nu}\theta = 0, \qquad \tau_\Pi D\Pi + \Pi + \zeta\theta + \delta_{\Pi\Pi}\Pi\theta = 0$$

(D, θ, $\Delta^{\mu\nu}$, $\Delta^{\mu\nu}_{\alpha\beta}$, $\sigma^{\mu\nu}$ as in the viscous section.) Charm, with $n = n(T, \alpha)$, $h = h(T)$, $P_0 = nT$ and $\delta M^{\mu\nu} = \pi_Q^{\mu\nu} + \Pi_Q \Delta^{\mu\nu}$:

$$\nabla_\mu (n\, u^\mu + \nu^\mu) = 0, \qquad \frac{D_s}{T}\, \Delta^\mu{}_\lambda \nabla_\nu T_Q^{\nu\lambda} + \nu^\mu = 0$$
$$T_Q^{\mu\nu} = (nh - P_0)\, u^\mu u^\nu + P_0\, \Delta^{\mu\nu} + h\, (u^\mu \nu^\nu + \nu^\mu u^\nu)$$
$$S^{\lambda\alpha\beta} = I_{31}\, (u^\lambda \Delta^{\alpha\beta} + 2 \text{ perm.}) + \mathcal{A}\, (u^\lambda u^\alpha \nu^\beta + 2 \text{ perm.}) + \mathcal{B}\, (\Delta^{\lambda\alpha}\nu^\beta + 2 \text{ perm.}) + \mathcal{C}\, (u^\lambda \delta M^{\alpha\beta} + 2 \text{ perm.})$$
$$\mathcal{E}^{\mu\nu} = \Delta^\mu{}_\alpha \Delta^\nu{}_\beta \nabla_\lambda S^{\lambda\alpha\beta}: \qquad \frac{D_s}{2T}\Big(\mathcal{E}^{\mu\nu} - \tfrac{1}{3}\Delta^{\mu\nu}\mathcal{E}^\lambda{}_\lambda\Big) + \pi_Q^{\mu\nu} = 0, \qquad \frac{D_s}{2T}\, \tfrac{1}{3}\mathcal{E}^\lambda{}_\lambda + \Pi_Q = 0$$

"2 perm." are the two other placements of the lone index. $I_{31}(T, \alpha)$, $\mathcal{A}(T)$, $\mathcal{B}(T)$, $\mathcal{C}(T)$ stay symbolic (Boltzmann: $I_{31} = nTh$, $\mathcal{A} = m^2 + 5Th$, $\mathcal{B} = Th$, $\mathcal{C} = mK_4/K_3$). The second moment δM is evolved but does not enter T_Q. In 2+1D the traceless rows also carry

$$-(1 - w_\omega)\, \tau_M W^{\mu\nu}, \qquad \tau_M = \frac{D_s \mathcal{C}}{2T}, \qquad W^{\mu\nu} = \pi_{Q\lambda}^\mu \omega^{\lambda\nu} + \pi_{Q\lambda}^\nu \omega^{\lambda\mu}, \qquad \omega^{\mu\nu} = \tfrac{1}{2}(\Delta^{\mu\lambda}\nabla_\lambda u^\nu - \Delta^{\nu\lambda}\nabla_\lambda u^\mu)$$

so $w_\omega = 1$ is the covariant row, which is exported; $w_\omega = 0$ removes the corotation coupling. The five systems (S1, S2 and every 0+1D system are derived for the checks; S3, S4, S5 in 1+1D and 2+1D are exported, with $\delta_{\Pi\Pi} = 0$):

$$\text{S1}: \ \nabla_\mu T_m^{\mu\nu} = 0, \ \pi = \Pi = 0 \qquad \text{S2}: \ \text{S1} + \text{the } \pi, \Pi \text{ equations}$$
$$\text{S3}: \ \text{charm number} + \text{first moment, background } T, u \qquad \text{S4}: \ \text{S3} + \text{second moment, background } T, u$$
$$\text{S5}: \ \nabla_\mu (T_m + T_Q)^{\mu\nu} = 0, \ \text{the } \pi, \Pi \text{ equations, the charm rows of S4; all fields evolved}$$

Conformal: $P = a_{\mathrm{SB}} T^4$, no bulk. Source on the left, $A_t\, \partial_\tau \phi + A_r\, \partial_r \phi + S = 0$ in 1+1D and $A_t\, \partial_\tau \phi + A_x\, \partial_x \phi + A_y\, \partial_y \phi + S = 0$ in 2+1D.

3.1.1 Fields

Medium, charm first moment, charm second moment, chart by chart:

$$0{+}1\text{D}: \ (T, \pi^\eta{}_\eta, \Pi), \ (\alpha), \ (\pi_Q{}^\eta{}_\eta, \Pi_Q)$$
$$1{+}1\text{D}: \ (T, u^r, \pi^\phi{}_\phi, \pi^\eta{}_\eta, \Pi), \ (\alpha, \nu^r), \ (\pi_Q{}^\phi{}_\phi, \pi_Q{}^\eta{}_\eta, \Pi_Q)$$

$$2{+}1\text{D}:\ (T,u^x,u^y,\pi^{yy},\pi^{\eta}{}_{\eta},\pi^{xy},\Pi),\ (\alpha,\nu^x,\nu^y),\ (\pi_Q^{yy},\pi_Q{}^{\eta}{}_{\eta},\pi_Q^{xy},\Pi_Q)$$

Rows, in the same order: the fluid rows $\nabla_\mu T^{\mu\tau}$, $\nabla_\mu T^{\mu i}$ (T_m+T_Q in S5); the shear rows $E^{\eta\eta}$ (0+1D), $E^{\phi\phi}$, $E^{\eta\eta}$ (1+1D), E^{yy}, $E^{\eta\eta}$, E^{xy} (2+1D); the bulk row; $\nabla_\mu N^\mu$; the current components; the traceless second-moment rows, components $\eta\eta$ (0+1D), $\phi\phi$, $\eta\eta$ (1+1D), yy, $\eta\eta$, xy (2+1D), as the medium's shear rows; the trace row.

In 0+1D the tensors are diagonal and $\nu=0$:

$$\pi^{\mu\nu}=\mathrm{diag}(0,-\frac{\pi^{\eta}{}_{\eta}}{2},-\frac{\pi^{\eta}{}_{\eta}}{2},\frac{\pi^{\eta}{}_{\eta}}{\tau^2}),\qquad \pi_Q^{\mu\nu}=\mathrm{diag}(0,-\frac{\pi_Q{}^{\eta}{}_{\eta}}{2},-\frac{\pi_Q{}^{\eta}{}_{\eta}}{2},\frac{\pi_Q{}^{\eta}{}_{\eta}}{\tau^2})$$

In 1+1D the medium $\pi^{\mu\nu}$ is as in the viscous section, and so is the charm stress: $\pi_Q{}^{\phi}{}_{\phi}$ and $\pi_Q{}^{\eta}{}_{\eta}$ are independent, the radial component follows from the trace. With the orthonormal triad transverse to u,

$$\pi_Q^{\mu\nu}=-(\pi_Q{}^{\phi}{}_{\phi}+\pi_Q{}^{\eta}{}_{\eta})\,\ell^\mu\ell^\nu+\pi_Q{}^{\phi}{}_{\phi}\,\hat\phi^\mu\hat\phi^\nu+\pi_Q{}^{\eta}{}_{\eta}\,\hat\eta^\mu\hat\eta^\nu$$
$$\ell^\mu=(u^r,u^\tau,0,0),\quad \hat\phi^\mu=(0,0,\frac{1}{r},0),\quad \hat\eta^\mu=(0,0,0,\frac{1}{\tau})$$

In 2+1D the medium $\pi^{\mu\nu}$ is as in the viscous section, and so is the charm stress: π_Q^{yy}, $\pi_Q{}^{\eta}{}_{\eta}$, π_Q^{xy} are independent, the τ row follows from $u_\mu\pi_Q^{\mu\nu}=0$ and

$$\pi_Q^{xx}=-\frac{-2\pi_Q^{xy}u^xu^y+\pi_Q^{yy}\,(u^x)^2+\pi_Q^{yy}+\pi_Q{}^{\eta}{}_{\eta}\,(u^x)^2+\pi_Q{}^{\eta}{}_{\eta}\,(u^y)^2+\pi_Q{}^{\eta}{}_{\eta}}{(u^y)^2+1}$$

3.2 Result

The 1+1D charm systems and the 1+1D conformal two-way system in full; the generic 1+1D two-way system and every 2+1D system as the pattern of nonzero entries (•), rows and columns in the field order given. Their entries are in the generated block of the file named with each function.

3.3 1+1D cylindrical

3.3.1 S3, charm first moment on a background

`one_d_HQ_BG_consistent_matrix` in `src/Matrix/1d_viscous_HQ_cilindrical_BG.jl`, fields $\phi=(\alpha,\nu^r)$.

$$(A_t)_{11}=(\partial_\alpha n)u^\tau$$
$$(A_t)_{12}=\frac{u^r}{u^\tau}$$
$$(A_t)_{21}=(\partial_\alpha n)D_s u^r u^\tau$$
$$(A_t)_{22}=\frac{D_s h u^\tau}{T}$$
$$(A_r)_{11}=(\partial_\alpha n)u^r$$
$$(A_r)_{12}=1$$
$$(A_r)_{21}=(\partial_\alpha n)D_s\left((u^r)^2+1\right)$$
$$(A_r)_{22}=\frac{D_s h u^r}{T}$$
$$\begin{aligned}S_1=&(\partial_r u^r)n+(\partial_T n)(\partial_r T)u^r+\frac{nu^r+\nu^r}{r}-\frac{(\partial_\tau u^r)\nu^r\,(u^r)^2}{(u^\tau)^3}+\frac{(\partial_\tau u^r)\nu^r}{u^\tau}+\frac{(\partial_\tau u^r)nu^r}{u^\tau}\\&+\frac{n\,(u^r)^2+n+\nu^r u^r}{\tau u^\tau}+(\partial_T n)(\partial_\tau T)u^\tau\end{aligned}$$
$$\begin{aligned}S_2=&\nu^r\\&+\frac{1}{Tu^\tau}D_s\Big(\left((u^r)^2+1\right)((\partial_T n)T+n)((\partial_r T)u^\tau+(\partial_\tau T)u^r)\\&+h'(T)\nu^r\left((\partial_r T)u^r u^\tau+(\partial_\tau T)\,(u^r)^2+(\partial_\tau T)\right)\Big)\end{aligned}$$

$$+\frac{1}{rT\tau\left(u^{\tau}\right)^{3}}D_{s}h\Big(\nu^{r}\tau u^{r}\left(u^{\tau}\right)^{3}+nr\tau\left(\left(u^{r}\right)^{2}+1\right)\left(\left(\partial_{r}u^{r}\right)u^{r}u^{\tau}+\left(\partial_{\tau}u^{r}\right)\left(u^{r}\right)^{2}+\left(\partial_{\tau}u^{r}\right)\right)$$
$$+\nu^{r}r\left(\left(u^{r}\right)^{2}\left(\left(\partial_{r}u^{r}\right)\tau u^{\tau}+2\right)+2(\partial_{r}u^{r})\tau u^{\tau}+\left(\partial_{\tau}u^{r}\right)\tau\left(\left(u^{r}\right)^{2}+1\right)u^{r}+\left(u^{r}\right)^{4}+1\right)\Big)$$

3.3.2 S4, charm first and second moment on a background

`one_d_HQ_BG_m2_matrix` in `src/Matrix/1d_viscous_HQ_cilindrical_BG_m2.jl`, fields $\phi=(\alpha,\nu^{r},{\pi_{Q}}^{\phi}{}_{\phi},{\pi_{Q}}^{\eta}{}_{\eta},\Pi_{Q})$.

$$(A_{t})_{11}=(\partial_{\alpha}n)u^{\tau}$$
$$(A_{t})_{12}=\frac{u^{r}}{u^{\tau}}$$
$$(A_{t})_{21}=(\partial_{\alpha}n)D_{s}u^{r}u^{\tau}$$
$$(A_{t})_{22}=\frac{D_{s}hu^{\tau}}{T}$$
$$(A_{t})_{32}=-\frac{\mathcal{B}D_{s}u^{r}}{3Tu^{\tau}}$$
$$(A_{t})_{33}=\frac{\mathcal{C}D_{s}u^{\tau}}{2T}$$
$$(A_{t})_{42}=-\frac{\mathcal{B}D_{s}u^{r}}{3Tu^{\tau}}$$
$$(A_{t})_{44}=\frac{\mathcal{C}D_{s}u^{\tau}}{2T}$$
$$(A_{t})_{51}=\frac{(\partial_{\alpha}I_{31})D_{s}u^{\tau}}{2T}$$
$$(A_{t})_{52}=\frac{5\mathcal{B}D_{s}u^{r}}{6Tu^{\tau}}$$
$$(A_{t})_{55}=\frac{\mathcal{C}D_{s}u^{\tau}}{2T}$$
$$(A_{r})_{11}=(\partial_{\alpha}n)u^{r}$$
$$(A_{r})_{12}=1$$
$$(A_{r})_{21}=(\partial_{\alpha}n)D_{s}\left(\left(u^{r}\right)^{2}+1\right)$$
$$(A_{r})_{22}=\frac{D_{s}hu^{r}}{T}$$
$$(A_{r})_{32}=-\frac{\mathcal{B}D_{s}}{3T}$$
$$(A_{r})_{33}=\frac{\mathcal{C}D_{s}u^{r}}{2T}$$
$$(A_{r})_{42}=-\frac{\mathcal{B}D_{s}}{3T}$$
$$(A_{r})_{44}=\frac{\mathcal{C}D_{s}u^{r}}{2T}$$
$$(A_{r})_{51}=\frac{(\partial_{\alpha}I_{31})D_{s}u^{r}}{2T}$$
$$(A_{r})_{52}=\frac{5\mathcal{B}D_{s}}{6T}$$
$$(A_{r})_{55}=\frac{\mathcal{C}D_{s}u^{r}}{2T}$$
$$S_{1}=(\partial_{r}u^{r})n+(\partial_{T}n)(\partial_{r}T)u^{r}+\frac{nu^{r}+\nu^{r}}{r}-\frac{(\partial_{\tau}u^{r})\nu^{r}\left(u^{r}\right)^{2}}{\left(u^{\tau}\right)^{3}}+\frac{(\partial_{\tau}u^{r})\nu^{r}}{u^{\tau}}+\frac{(\partial_{\tau}u^{r})nu^{r}}{u^{\tau}}$$
$$+\frac{n\left(u^{r}\right)^{2}+n+\nu^{r}u^{r}}{\tau u^{\tau}}+(\partial_{T}n)(\partial_{\tau}T)u^{\tau}$$
$$S_{2}=\nu^{r}$$
$$+\frac{1}{Tu^{\tau}}D_{s}\Big(\left(\left(u^{r}\right)^{2}+1\right)((\partial_{T}n)T+n)((\partial_{r}T)u^{\tau}+(\partial_{\tau}T)u^{r})$$

$$
\begin{aligned}
&+ h'(T)\nu^r \left((\partial_r T) u^r u^\tau + (\partial_\tau T) \left(u^r\right)^2 + (\partial_\tau T) \right) \Big) \\
&+ \frac{1}{rT\tau \left(u^\tau\right)^3} D_s h \Big(\nu^r \tau u^r \left(u^\tau\right)^3 + nr\tau \left(\left(u^r\right)^2 + 1 \right) \left((\partial_r u^r) u^r u^\tau + (\partial_\tau u^r) \left(u^r\right)^2 + (\partial_\tau u^r) \right) \\
&+ \nu^r r \left(\left(u^r\right)^2 \left((\partial_r u^r) \tau u^\tau + 2 \right) + 2 (\partial_r u^r) \tau u^\tau + (\partial_\tau u^r) \tau \left(\left(u^r\right)^2 + 1 \right) u^r + \left(u^r\right)^4 + 1 \right) \Big) \\
S_3 = {} & \pi_Q{}^\phi{}_\phi \\
&- \frac{1}{6rT\tau \left(u^\tau\right)^3} D_s \Big(2\mathcal{A} (\partial_\tau u^r) \nu^r r\tau + 2\mathcal{B} (\partial_\tau u^r) \nu^r r\tau - 3\mathcal{C}'(T) (\partial_\tau T) \pi_Q{}^\phi{}_\phi r\tau + 2\mathcal{B} \nu^r r u^r \\
&+ 2\mathcal{B}'(T) (\partial_\tau T) \nu^r r\tau u^r + 2\mathcal{A} (\partial_\tau u^r) \nu^r r\tau \left(u^r\right)^2 - 6\mathcal{C}'(T) (\partial_\tau T) \pi_Q{}^\phi{}_\phi r\tau \left(u^r\right)^2 \\
&+ 2\mathcal{B} \nu^r r \left(u^r\right)^3 + 2\mathcal{B}'(T) (\partial_\tau T) \nu^r r\tau \left(u^r\right)^3 - 3\mathcal{C}'(T) (\partial_\tau T) \pi_Q{}^\phi{}_\phi r\tau \left(u^r\right)^4 - 4\mathcal{B} \nu^r \tau u^\tau \\
&+ 2\mathcal{B}'(T) (\partial_r T) \nu^r r\tau u^\tau + 2\mathcal{A} (\partial_r u^r) \nu^r r\tau u^r u^\tau - 3\mathcal{C}'(T) (\partial_r T) \pi_Q{}^\phi{}_\phi r\tau u^r u^\tau - 4\mathcal{B} \nu^r \tau \left(u^r\right)^2 u^\tau \\
&+ 2\mathcal{B}'(T) (\partial_r T) \nu^r r\tau \left(u^r\right)^2 u^\tau - 3\mathcal{C}'(T) (\partial_r T) \pi_Q{}^\phi{}_\phi r\tau \left(u^r\right)^3 u^\tau \\
&+ 2I_{31} \left(\left(u^r\right)^2 + 1 \right) \left(r \left((\partial_r u^r) \tau u^\tau + (\partial_\tau u^r) \tau u^r + \left(u^r\right)^2 + 1 \right) - 2\tau u^r u^\tau \right) \\
&+ \mathcal{C} \left(\left(u^r\right)^2 + 1 \right) \Big(-4\Pi_Q \tau u^r u^\tau - 7\pi_Q{}^\phi{}_\phi \tau u^r u^\tau - 2\pi_Q{}^\eta{}_\eta r \left((\partial_r u^r) \tau u^\tau + (\partial_\tau u^r) \tau u^r - \left(u^r\right)^2 - 1 \right) \\
&+ 2\Pi_Q r \left((\partial_r u^r) \tau u^\tau + (\partial_\tau u^r) \tau u^r + \left(u^r\right)^2 + 1 \right) \\
&- \pi_Q{}^\phi{}_\phi r \left(5 (\partial_r u^r) \tau u^\tau + 5 (\partial_\tau u^r) \tau u^r + 3 \left(u^r\right)^2 + 3 \right) \Big) \Big) \\
S_4 = {} & \pi_Q{}^\eta{}_\eta \\
&+ \frac{1}{6rT\tau \left(u^\tau\right)^3} D_s \Big(-2\mathcal{A} (\partial_\tau u^r) \nu^r r\tau - 2\mathcal{B} (\partial_\tau u^r) \nu^r r\tau + 3\mathcal{C}'(T) (\partial_\tau T) \pi_Q{}^\eta{}_\eta r\tau + 4\mathcal{B} \nu^r r u^r \\
&- 2\mathcal{B}'(T) (\partial_\tau T) \nu^r r\tau u^r - 2\mathcal{A} (\partial_\tau u^r) \nu^r r\tau \left(u^r\right)^2 + 6\mathcal{C}'(T) (\partial_\tau T) \pi_Q{}^\eta{}_\eta r\tau \left(u^r\right)^2 \\
&+ 4\mathcal{B} \nu^r r \left(u^r\right)^3 - 2\mathcal{B}'(T) (\partial_\tau T) \nu^r r\tau \left(u^r\right)^3 + 3\mathcal{C}'(T) (\partial_\tau T) \pi_Q{}^\eta{}_\eta r\tau \left(u^r\right)^4 - 2\mathcal{B} \nu^r \tau u^\tau \\
&- 2\mathcal{B}'(T) (\partial_r T) \nu^r r\tau u^\tau - 2\mathcal{A} (\partial_r u^r) \nu^r r\tau u^r u^\tau + 3\mathcal{C}'(T) (\partial_r T) \pi_Q{}^\eta{}_\eta r\tau u^r u^\tau - 2\mathcal{B} \nu^r \tau \left(u^r\right)^2 u^\tau \\
&- 2\mathcal{B}'(T) (\partial_r T) \nu^r r\tau \left(u^r\right)^2 u^\tau + 3\mathcal{C}'(T) (\partial_r T) \pi_Q{}^\eta{}_\eta r\tau \left(u^r\right)^3 u^\tau \\
&- 2I_{31} \left(\left(u^r\right)^2 + 1 \right) \left(r \left((\partial_r u^r) \tau u^\tau + (\partial_\tau u^r) \tau u^r - 2 \left(u^r\right)^2 - 2 \right) + \tau u^r u^\tau \right) \\
&+ \mathcal{C} \left(\left(u^r\right)^2 + 1 \right) \Big(2\pi_Q{}^\phi{}_\phi \tau (u^\tau ((\partial_r u^r) r - u^r) + (\partial_\tau u^r) r u^r) \\
&- 2\Pi_Q \left(r \left((\partial_r u^r) \tau u^\tau + (\partial_\tau u^r) \tau u^r - 2 \left(u^r\right)^2 - 2 \right) + \tau u^r u^\tau \right) \\
&+ \pi_Q{}^\eta{}_\eta \left(r \left(5 (\partial_r u^r) \tau u^\tau + 5 (\partial_\tau u^r) \tau u^r + 7 \left(u^r\right)^2 + 7 \right) + 3\tau u^r u^\tau \right) \Big) \Big) \\
S_5 = {} & \Pi_Q \\
&+ \frac{1}{6rT\tau \left(u^\tau\right)^3} D_s \Big(3 (\partial_T I_{31}) (\partial_\tau T) r\tau + 2\mathcal{A} (\partial_\tau u^r) \nu^r r\tau + 5\mathcal{B} (\partial_\tau u^r) \nu^r r\tau + 3\mathcal{C}'(T) (\partial_\tau T) \Pi_Q r\tau + 5\mathcal{B} \nu^r r u^r \\
&+ 5\mathcal{B}'(T) (\partial_\tau T) \nu^r r\tau u^r + 6 (\partial_T I_{31}) (\partial_\tau T) r\tau \left(u^r\right)^2 + 2\mathcal{A} (\partial_\tau u^r) \nu^r r\tau \left(u^r\right)^2 \\
&+ 6\mathcal{C}'(T) (\partial_\tau T) \Pi_Q r\tau \left(u^r\right)^2 + 5\mathcal{B} \nu^r r \left(u^r\right)^3 + 5\mathcal{B}'(T) (\partial_\tau T) \nu^r r\tau \left(u^r\right)^3 \\
&+ 3 (\partial_T I_{31}) (\partial_\tau T) r\tau \left(u^r\right)^4 + 3\mathcal{C}'(T) (\partial_\tau T) \Pi_Q r\tau \left(u^r\right)^4 + 5\mathcal{B} \nu^r \tau u^\tau + 5\mathcal{B}'(T) (\partial_r T) \nu^r r\tau u^\tau \\
&+ 3 (\partial_T I_{31}) (\partial_r T) r\tau u^r u^\tau + 2\mathcal{A} (\partial_r u^r) \nu^r r\tau u^r u^\tau + 3\mathcal{C}'(T) (\partial_r T) \Pi_Q r\tau u^r u^\tau + 5\mathcal{B} \nu^r \tau \left(u^r\right)^2 u^\tau \\
&+ 5\mathcal{B}'(T) (\partial_r T) \nu^r r\tau \left(u^r\right)^2 u^\tau + 3 (\partial_T I_{31}) (\partial_r T) r\tau \left(u^r\right)^3 u^\tau + 3\mathcal{C}'(T) (\partial_r T) \Pi_Q r\tau \left(u^r\right)^3 u^\tau \\
&+ 5I_{31} \left(\left(u^r\right)^2 + 1 \right) \left(r \left((\partial_r u^r) \tau u^\tau + (\partial_\tau u^r) \tau u^r + \left(u^r\right)^2 + 1 \right) + \tau u^r u^\tau \right) \\
&+ \mathcal{C} \left(\left(u^r\right)^2 + 1 \right) \Big(5\Pi_Q \left(r \left((\partial_r u^r) \tau u^\tau + (\partial_\tau u^r) \tau u^r + \left(u^r\right)^2 + 1 \right) + \tau u^r u^\tau \right) \\
&- 2 \Big(\pi_Q{}^\eta{}_\eta r \left((\partial_r u^r) \tau u^\tau + (\partial_\tau u^r) \tau u^r - \left(u^r\right)^2 - 1 \right) \\
&+ \pi_Q{}^\phi{}_\phi \tau (u^\tau ((\partial_r u^r) r - u^r) + (\partial_\tau u^r) r u^r) \Big) \Big) \Big)
\end{aligned}
$$

3.3.3 S5, two-way

`one_d_two_way_matrix` in `src/Matrix/1d_two_way_cilindrical.jl`, fields $\phi = (T, u^r, \pi^\phi{}_\phi, \pi^\eta{}_\eta, \Pi, \alpha, \nu^r, \pi_Q{}^\phi{}_\phi, \pi_Q{}^\eta{}_\eta, \Pi_Q)$.

$$A_t = \begin{pmatrix}
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet
\end{pmatrix}, \qquad A_r = \begin{pmatrix}
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet
\end{pmatrix},$$

$$S = \begin{pmatrix}
\bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet
\end{pmatrix}$$

3.3.4 S5, two-way, conformal

`one_d_two_way_matrix_conformal` in `src/Matrix/1d_two_way_cilindrical.jl`, fields $\phi = (T, u^r, \pi^\phi{}_\phi, \pi^\eta{}_\eta, \alpha, \nu^r, \pi_Q{}^\phi{}_\phi, \pi_Q$

$$\begin{aligned}
(A_t)_{11} &= 2h'(T)\nu^r u^r + 4a_{\mathrm{SB}}T^3\left(4\left(u^r\right)^2 + 3\right) + n\left(h'(T)\left(u^r\right)^2 + h'(T) - 1\right) \\
&\quad + (\partial_T n)\left(h\left(u^r\right)^2 + h - T\right) \\
(A_t)_{12} &= 2h(nu^r + \nu^r) - 2u^r\left(-4a_{\mathrm{SB}}T^4 + \pi^\eta{}_\eta + \pi^\phi{}_\phi\right) \\
(A_t)_{13} &= -\left(u^r\right)^2 \\
(A_t)_{14} &= -\left(u^r\right)^2 \\
(A_t)_{15} &= (\partial_\alpha n)\left(h\left(u^r\right)^2 + h - T\right) \\
(A_t)_{16} &= 2hu^r \\
(A_t)_{21} &= \frac{u^r\left(\left(u^r\right)^2 + 1\right)\left(16a_{\mathrm{SB}}T^3 + (\partial_T n)h\right) + h'(T)\left(n\left(\left(u^r\right)^2 + 1\right)u^r + 2\nu^r\left(u^r\right)^2 + \nu^r\right)}{u^\tau} \\
(A_t)_{22} &= \frac{1}{\left(u^\tau\right)^3}\Big(-\left(2\left(u^r\right)^4 + 3\left(u^r\right)^2 + 1\right)\left(-4a_{\mathrm{SB}}T^4 + \pi^\eta{}_\eta + \pi^\phi{}_\phi\right) \\
&\quad + h\left(2n\left(u^r\right)^4 + 3n\left(u^r\right)^2 + n + 2\nu^r\left(u^r\right)^3 + 3\nu^r u^r\right)\Big) \\
(A_t)_{23} &= -u^r u^\tau \\
(A_t)_{24} &= -u^r u^\tau \\
(A_t)_{25} &= (\partial_\alpha n)hu^r u^\tau \\
(A_t)_{26} &= \frac{h\left(2\left(u^r\right)^2 + 1\right)}{u^\tau} \\
(A_t)_{32} &= \frac{u^r(3\delta_{\pi\pi}\pi^\phi{}_\phi - 2\eta)}{3r^2u^\tau} \\
(A_t)_{33} &= \frac{\tau_\pi u^\tau}{r^2} \\
(A_t)_{42} &= \frac{u^r(3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta)}{3\tau^2u^\tau}
\end{aligned}$$

$$(A_t)_{44} = \frac{\tau_\pi u^\tau}{\tau^2}$$
$$(A_t)_{51} = (\partial_T n) u^\tau$$
$$(A_t)_{52} = \frac{n u^r \left((u^r)^2 + 1 \right) + \nu^r}{(u^\tau)^3}$$
$$(A_t)_{55} = (\partial_\alpha n) u^\tau$$
$$(A_t)_{56} = \frac{u^r}{u^\tau}$$
$$(A_t)_{61} = \frac{D_s u^\tau (h'(T)\nu^r + u^r((\partial_T n)T + n))}{T}$$
$$(A_t)_{62} = \frac{D_s h \left(n (u^r)^2 + n + \nu^r u^r \right)}{T u^\tau}$$
$$(A_t)_{65} = (\partial_\alpha n) D_s u^r u^\tau$$
$$(A_t)_{66} = \frac{D_s h u^\tau}{T}$$
$$(A_t)_{71} = \frac{D_s \left(3\mathcal{C}'(T) {\pi_Q}^\phi{}_\phi \left((u^r)^2 + 1 \right) - 2\mathcal{B}'(T)\nu^r u^r \right)}{6T u^\tau}$$
$$(A_t)_{72} = -\frac{D_s \left(2\mathcal{A}\nu^r \left((u^r)^2 + 1 \right) + 2\mathcal{B}\nu^r + u^r \left((u^r)^2 + 1 \right) (2\mathcal{C}\Pi_Q - 2\mathcal{C}{\pi_Q}^\eta{}_\eta - 5\mathcal{C}{\pi_Q}^\phi{}_\phi + 2I_{31}) \right)}{6T (u^\tau)^3}$$
$$(A_t)_{76} = -\frac{\mathcal{B} D_s u^r}{3T u^\tau}$$
$$(A_t)_{77} = \frac{\mathcal{C} D_s u^\tau}{2T}$$
$$(A_t)_{81} = \frac{D_s \left(3\mathcal{C}'(T) {\pi_Q}^\eta{}_\eta \left((u^r)^2 + 1 \right) - 2\mathcal{B}'(T)\nu^r u^r \right)}{6T u^\tau}$$
$$(A_t)_{82} = -\frac{D_s \left(2\mathcal{A}\nu^r \left((u^r)^2 + 1 \right) + 2\mathcal{B}\nu^r + u^r \left((u^r)^2 + 1 \right) (2\mathcal{C}\Pi_Q - 5\mathcal{C}{\pi_Q}^\eta{}_\eta - 2\mathcal{C}{\pi_Q}^\phi{}_\phi + 2I_{31}) \right)}{6T (u^\tau)^3}$$
$$(A_t)_{86} = -\frac{\mathcal{B} D_s u^r}{3T u^\tau}$$
$$(A_t)_{88} = \frac{\mathcal{C} D_s u^\tau}{2T}$$
$$(A_t)_{91} = \frac{D_s \left(5\mathcal{B}'(T)\nu^r u^r + 3\mathcal{C}'(T)\Pi_Q \left((u^r)^2 + 1 \right) + 3(\partial_T I_{31}) \left((u^r)^2 + 1 \right) \right)}{6T u^\tau}$$
$$(A_t)_{92} = \frac{D_s \left(2\mathcal{A}\nu^r \left((u^r)^2 + 1 \right) + 5\mathcal{B}\nu^r + u^r \left((u^r)^2 + 1 \right) (5\mathcal{C}\Pi_Q - 2\mathcal{C}({\pi_Q}^\eta{}_\eta + {\pi_Q}^\phi{}_\phi) + 5I_{31}) \right)}{6T (u^\tau)^3}$$
$$(A_t)_{95} = \frac{(\partial_\alpha I_{31}) D_s u^\tau}{2T}$$
$$(A_t)_{96} = \frac{5\mathcal{B} D_s u^r}{6T u^\tau}$$
$$(A_t)_{99} = \frac{\mathcal{C} D_s u^\tau}{2T}$$
$$(A_r)_{11} = \frac{u^r \left((u^r)^2 + 1 \right) \left(16 a_{\mathrm{SB}} T^3 + (\partial_T n) h \right) + h'(T) \left(n \left((u^r)^2 + 1 \right) u^r + 2\nu^r (u^r)^2 + \nu^r \right)}{u^\tau}$$
$$(A_r)_{12} = \frac{1}{(u^\tau)^3} \Big(- \left(2 (u^r)^4 + 3 (u^r)^2 + 1 \right) \left(-4 a_{\mathrm{SB}} T^4 + {\pi^\eta}_\eta + {\pi^\phi}_\phi \right) + h \left(2n (u^r)^4 + 3n (u^r)^2 + n + 2\nu^r (u^r)^3 + 3\nu^r u^r \right) \Big)$$
$$(A_r)_{13} = -u^r u^\tau$$
$$(A_r)_{14} = -u^r u^\tau$$
$$(A_r)_{15} = (\partial_\alpha n) h u^r u^\tau$$

$$(A_r)_{16} = \frac{h\left(2\left(u^r\right)^2+1\right)}{u^\tau}$$
$$(A_r)_{21} = 16a_{\rm SB}T^3\left(u^r\right)^2 + 4a_{\rm SB}T^3 + h'(T)n\left(u^r\right)^2 + 2h'(T)\nu^r u^r + (\partial_T n)h\left(u^r\right)^2 + (\partial_T n)T + n$$
$$(A_r)_{22} = 2h(nu^r+\nu^r) - 2u^r\left(-4a_{\rm SB}T^4 + \pi^\eta{}_\eta + \pi^\phi{}_\phi\right)$$
$$(A_r)_{23} = -\left(u^r\right)^2 - 1$$
$$(A_r)_{24} = -\left(u^r\right)^2 - 1$$
$$(A_r)_{25} = (\partial_\alpha n)\left(h\left(u^r\right)^2 + T\right)$$
$$(A_r)_{26} = 2hu^r$$
$$(A_r)_{32} = \frac{3\delta_{\pi\pi}\pi^\phi{}_\phi - 2\eta}{3r^2}$$
$$(A_r)_{33} = \frac{\tau_\pi u^r}{r^2}$$
$$(A_r)_{42} = \frac{3\delta_{\pi\pi}\pi^\eta{}_\eta - 2\eta}{3\tau^2}$$
$$(A_r)_{44} = \frac{\tau_\pi u^r}{\tau^2}$$
$$(A_r)_{51} = (\partial_T n)u^r$$
$$(A_r)_{52} = n$$
$$(A_r)_{55} = (\partial_\alpha n)u^r$$
$$(A_r)_{56} = 1$$
$$(A_r)_{61} = \frac{D_s\left(h'(T)\nu^r u^r + (\partial_T n)T\left(\left(u^r\right)^2+1\right) + n\left(\left(u^r\right)^2+1\right)\right)}{T}$$
$$(A_r)_{62} = \frac{D_s h\left(nu^r\left(\left(u^r\right)^2+1\right) + \nu^r\left(\left(u^r\right)^2+2\right)\right)}{T\left(u^\tau\right)^2}$$
$$(A_r)_{65} = (\partial_\alpha n)D_s\left(\left(u^r\right)^2+1\right)$$
$$(A_r)_{66} = \frac{D_s h u^r}{T}$$
$$(A_r)_{71} = -\frac{D_s(2\mathcal{B}'(T)\nu^r - 3\mathcal{C}'(T)\pi_Q{}^\phi{}_\phi u^r)}{6T}$$
$$(A_r)_{72} = -\frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(\left(u^r\right)^2+1\right)(2\Pi_Q - 2\pi_Q{}^\eta{}_\eta - 5\pi_Q{}^\phi{}_\phi) + 2I_{31}\left(\left(u^r\right)^2+1\right)\right)}{6T\left(u^\tau\right)^2}$$
$$(A_r)_{76} = -\frac{\mathcal{B}D_s}{3T}$$
$$(A_r)_{77} = \frac{\mathcal{C}D_s u^r}{2T}$$
$$(A_r)_{81} = -\frac{D_s(2\mathcal{B}'(T)\nu^r - 3\mathcal{C}'(T)\pi_Q{}^\eta{}_\eta u^r)}{6T}$$
$$(A_r)_{82} = -\frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(\left(u^r\right)^2+1\right)(2\Pi_Q - 5\pi_Q{}^\eta{}_\eta - 2\pi_Q{}^\phi{}_\phi) + 2I_{31}\left(\left(u^r\right)^2+1\right)\right)}{6T\left(u^\tau\right)^2}$$
$$(A_r)_{86} = -\frac{\mathcal{B}D_s}{3T}$$
$$(A_r)_{88} = \frac{\mathcal{C}D_s u^r}{2T}$$
$$(A_r)_{91} = \frac{D_s(5\mathcal{B}'(T)\nu^r + 3u^r(\mathcal{C}'(T)\Pi_Q + (\partial_T I_{31})))}{6T}$$
$$(A_r)_{92} = \frac{D_s\left(2\mathcal{A}\nu^r u^r + \mathcal{C}\left(\left(u^r\right)^2+1\right)(5\Pi_Q - 2(\pi_Q{}^\eta{}_\eta + \pi_Q{}^\phi{}_\phi)) + 5I_{31}\left(\left(u^r\right)^2+1\right)\right)}{6T\left(u^\tau\right)^2}$$
$$(A_r)_{95} = \frac{(\partial_\alpha I_{31})D_s u^r}{2T}$$

$$
\begin{aligned}
(A_r)_{96} &= \frac{5\mathcal{B}D_s}{6T} \\
(A_r)_{99} &= \frac{\mathcal{C}D_s u^r}{2T} \\
S_1 &= \frac{1}{r\tau u^\tau}\Big(-\pi^\phi{}_\phi \tau u^r + 4a_{\rm SB}T^4\tau u^r - \pi^\phi{}_\phi \tau \left(u^r\right)^3 + 4a_{\rm SB}T^4\tau \left(u^r\right)^3 + 4a_{\rm SB}rT^4u^\tau - \pi^\phi{}_\phi r \left(u^r\right)^2 u^\tau \\
&\quad + 4a_{\rm SB}rT^4 \left(u^r\right)^2 u^\tau - \pi^\eta{}_\eta \left(r\left(\left(u^r\right)^2 - 1\right)u^\tau + \tau\left(\left(u^r\right)^3 + u^r\right)\right) \\
&\quad + h\left(n\left(\left(u^r\right)^2+1\right)\left(ru^\tau + \tau u^r\right) + \nu^r\left(2ru^ru^\tau + 2\tau\left(u^r\right)^2 + \tau\right)\right)\Big) \\
S_2 &= \frac{1}{r\tau u^\tau}\Big(-\pi^\phi{}_\phi r u^r + 4a_{\rm SB}rT^4u^r - \pi^\phi{}_\phi r\left(u^r\right)^3 + 4a_{\rm SB}rT^4\left(u^r\right)^3 - 2\pi^\phi{}_\phi\tau u^\tau - \pi^\phi{}_\phi\tau\left(u^r\right)^2u^\tau \\
&\quad + 4a_{\rm SB}T^4\tau\left(u^r\right)^2u^\tau - \pi^\eta{}_\eta\left(\left(u^r\right)^2+1\right)\left(ru^r+\tau u^\tau\right) + hnu^r\left(r\left(u^r\right)^2 + r + \tau u^ru^\tau\right) \\
&\quad + h\nu^r\left(2r\left(u^r\right)^2 + r + 2\tau u^ru^\tau\right)\Big) \\
S_3 &= \frac{3\pi^\phi{}_\phi(r(\delta_{\pi\pi}u^\tau + \tau) + \delta_{\pi\pi}\tau u^r) - 2ru^\tau\eta + 4\tau u^r\eta}{3r^3\tau} \\
S_4 &= \frac{3\pi^\eta{}_\eta(r(\delta_{\pi\pi}u^\tau + \tau) + \delta_{\pi\pi}\tau u^r) + 4ru^\tau\eta - 2\tau u^r\eta}{3r\tau^3} \\
S_5 &= \frac{nu^r + \nu^r}{r} + \frac{n\left(u^r\right)^2 + n + \nu^ru^r}{\tau u^\tau} \\
S_6 &= \frac{D_sh\nu^r\left(r\left(u^r\right)^2 + r + \tau u^ru^\tau\right)}{rT\tau u^\tau} + \nu^r \\
S_7 &= \pi_Q{}^\phi{}_\phi \\
&\quad + \frac{1}{6rT\tau u^\tau}D_s\Big(-2\mathcal{B}\nu^r(ru^r - 2\tau u^\tau) - 2I_{31}\left(r\left(u^r\right)^2 + r - 2\tau u^ru^\tau\right) \\
&\quad + \mathcal{C}\Big(-2\pi_Q{}^\eta{}_\eta r\left(\left(u^r\right)^2+1\right) + 3\pi_Q{}^\phi{}_\phi r\left(\left(u^r\right)^2+1\right) + 7\pi_Q{}^\phi{}_\phi\tau u^ru^\tau \\
&\quad - 2\Pi_Q\left(r\left(u^r\right)^2 + r - 2\tau u^ru^\tau\right)\Big)\Big) \\
S_8 &= \pi_Q{}^\eta{}_\eta \\
&\quad + \frac{1}{6rT\tau u^\tau}D_s\Big(4I_{31}r\left(\left(u^r\right)^2+1\right) - 2I_{31}\tau u^ru^\tau + 2\mathcal{B}\nu^r(2ru^r - \tau u^\tau) \\
&\quad + \mathcal{C}\left(4\Pi_Qr\left(\left(u^r\right)^2+1\right) - 2\Pi_Q\tau u^ru^\tau + 7\pi_Q{}^\eta{}_\eta r\left(\left(u^r\right)^2+1\right) + 3\pi_Q{}^\eta{}_\eta\tau u^ru^\tau - 2\pi_Q{}^\phi{}_\phi\tau u^ru^\tau\right)\Big) \\
S_9 &= \Pi_Q \\
&\quad + \frac{1}{6rT\tau u^\tau}D_s\Big(5\mathcal{B}\nu^r(ru^\tau + \tau u^\tau) + 5I_{31}\left(r\left(u^r\right)^2 + r + \tau u^ru^\tau\right) \\
&\quad + \mathcal{C}\left(5\Pi_Q\left(r\left(u^r\right)^2 + r + \tau u^ru^\tau\right) + 2\pi_Q{}^\eta{}_\eta r\left(\left(u^r\right)^2+1\right) + 2\pi_Q{}^\phi{}_\phi\tau u^ru^\tau\right)\Big)
\end{aligned}
$$

3.4 2+1D Cartesian

3.4.1 S3, charm first moment on a background

`two_d_HQ_BG_consistent_matrix` in `src/Matrix/2d_viscous_HQ_BG.jl`, fields $\phi = (\alpha, \nu^x, \nu^y)$.

$$
A_t = \begin{pmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \cdot \\ \bullet & \cdot & \bullet \end{pmatrix}, \qquad A_x = \begin{pmatrix} \bullet & \bullet & \cdot \\ \bullet & \bullet & \cdot \\ \bullet & \cdot & \bullet \end{pmatrix}, \qquad A_y = \begin{pmatrix} \bullet & \cdot & \bullet \\ \bullet & \bullet & \cdot \\ \bullet & \cdot & \bullet \end{pmatrix}, \qquad S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \end{pmatrix}
$$

3.4.2 S4, charm first and second moment on a background

`two_d_HQ_BG_m2_matrix` in `src/Matrix/2d_viscous_HQ_BG_m2.jl`, fields $\phi = (\alpha, \nu^x, \nu^y, \pi_Q^{yy}, \pi_Q{}^\eta{}_\eta, \pi_Q^{xy}, \Pi_Q)$.

$$A_t = \begin{pmatrix} \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_x = \begin{pmatrix} \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_y = \begin{pmatrix} \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\ \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}$$

$$S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{pmatrix}$$

3.4.3 S5, two-way

`two_d_two_way_matrix` in `src/Matrix/2d_two_way.jl`, fields $\phi = (T, u^x, u^y, \pi^{yy}, \pi^{\eta}{}_{\eta}, \pi^{xy}, \Pi, \alpha, \nu^x, \nu^y, \pi_Q^{yy}, \pi_Q{}^{\eta}{}_{\eta}, \pi_Q^{xy}, \Pi_Q)$.

$$A_t = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}, \qquad A_x = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet \end{pmatrix}$$

$$A_y = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\ \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet \end{pmatrix}, \qquad S = \begin{pmatrix} \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \end{pmatrix}$$

3.4.4 S5, two-way, conformal

`two_d_two_way_matrix_conformal` in `src/Matrix/2d_two_way.jl`, fields $\phi = (T, u^x, u^y, \pi^{yy}, \pi^{\eta}{}_{\eta}, \pi^{xy}, \alpha, \nu^x, \nu^y, \pi_Q^{yy}, \pi_Q{}^{\eta}{}_{\eta}, \pi$

$$A_t = \begin{pmatrix}
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \bullet
\end{pmatrix}, \qquad A_x = \begin{pmatrix}
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \bullet & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet
\end{pmatrix}$$

$$A_y = \begin{pmatrix}
\bullet & \bullet & \bullet & \bullet & \cdot & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \bullet & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \bullet & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\bullet & \cdot & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \cdot & \bullet & \cdot & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \cdot & \bullet & \bullet & \cdot & \bullet & \cdot \\
\bullet & \bullet & \bullet & \cdot & \cdot & \cdot & \bullet & \cdot & \bullet & \cdot & \cdot & \bullet
\end{pmatrix}, \qquad S = \begin{pmatrix}
\bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet \\ \bullet
\end{pmatrix}$$

3.5 Checks

id	what it tests	
`G0-0p1d`	Nabla g == 0	PASS
`G1-0p1d`	trap detector: Nabla.(X(tau,..),0,0,0) == X' + X/tau (Christoffel AND sqrt(-g) routes)	PASS
`G2-0p1d`	u.u=-1; u.pi=0, tr pi=0 (medium); u.nu=0; u.piQ=0, tr piQ=0; both symmetric	PASS
`G-aQ-0p1d`	aQ' has identically vanishing coefficient in the second-moment rows (symbolic); dropped	PASS
`G3-0p1d`	every row (medium, charm-in-T, first moment, second moment) is quasi-linear	PASS
`S3-free-0p1d`	S3 rows carry no second-moment field	PASS
`S34-nomedpi-0p1d`	charm rows never see the medium pi or Pi (so S3/S4 need only T, u as background)	PASS
`Gc-extract-0p1d`	A_t dphi + A_x dphi (+A_y dphi) + src == the raw rows (symbolic)	PASS
`Gc-ctrl-0p1d`	control: the exact random-point zero test REJECTS a 1e-9-perturbed identity and a nonzero row	PASS
`Gc-total-0p1d`	S5 fluid rows == Nabla_mu (T_m + T_Q)^{mu nu}, sqrt(-g) route (symbolic)	PASS
`Gc-number-0p1d`	charm row == Nabla.N, sqrt(-g) route (symbolic)	PASS
`Gc-dM-0p1d`	the second moment is passive: dM columns of the fluid + first-moment rows vanish identically, and their sources carry no dM	PASS
`Gc-off-0p1d`	charm off (n, nu, dM -> 0): S5's medium block and medium sources = = S2 exactly; the charm rows' medium columns and sources vanish, so charm = 0 is an invariant subspace on which S5 IS S2	PASS
`G0-1p1d`	Nabla g == 0	PASS
`G1-1p1d`	trap detector: Nabla.(X(tau,..),0,0,0) == X' + X/tau (Christoffel AND sqrt(-g) routes)	PASS
`G2-1p1d`	u.u=-1; u.pi=0, tr pi=0 (medium); u.nu=0; u.piQ=0, tr piQ=0; both symmetric	PASS
`G2-triad`	the (l, phihat, etahat) triad is orthonormal and transverse	PASS

id	what it tests	
G-aQ-1p1d	aQ' has identically vanishing coefficient in the second-moment rows (symbolic); dropped	PASS
G3-1p1d	every row (medium, charm-in-T, first moment, second moment) is quasi-linear	PASS
S3-free-1p1d	S3 rows carry no second-moment field	PASS
S34-nomedpi-1p1d	charm rows never see the medium pi or Pi (so S3/S4 need only T, u as background)	PASS
Gc-extract-1p1d	A_t dphi + A_x dphi (+A_y dphi) + src == the raw rows (symbolic)	PASS
Gc-ctrl-1p1d	control: the exact random-point zero test REJECTS a 1e-9-perturbed identity and a nonzero row	PASS
Gc-total-1p1d	S5 fluid rows == Nabla_mu (T_m + T_Q)^{mu nu}, sqrt(-g) route (symbolic)	PASS
Gc-number-1p1d	charm row == Nabla.N, sqrt(-g) route (symbolic)	PASS
Gc-dM-1p1d	the second moment is passive: dM columns of the fluid + first-moment rows vanish identically, and their sources carry no dM	PASS
Gc-off-1p1d	charm off (n, nu, dM -> 0): S5's medium block and medium sources = = S2 exactly; the charm rows' medium columns and sources vanish, so charm = 0 is an invariant subspace on which S5 IS S2	PASS
G0-2p1d	Nabla g == 0	PASS
G1-2p1d	trap detector: Nabla.(X(tau,..),0,0,0) == X' + X/tau (Christoffel AND sqrt(-g) routes)	PASS
G2-2p1d	u.u=-1; u.pi=0, tr pi=0 (medium); u.nu=0; u.piQ=0, tr piQ=0; both symmetric	PASS
G-aQ-2p1d	aQ' has identically vanishing coefficient in the second-moment rows (symbolic); dropped	PASS
G3-2p1d	every row (medium, charm-in-T, first moment, second moment) is quasi-linear	PASS
S3-free-2p1d	S3 rows carry no second-moment field	PASS
S34-nomedpi-2p1d	charm rows never see the medium pi or Pi (so S3/S4 need only T, u as background)	PASS
Gc-extract-2p1d	A_t dphi + A_x dphi (+A_y dphi) + src == the raw rows (exact at 4 random rational points)	PASS
Gc-ctrl-2p1d	control: the exact random-point zero test REJECTS a 1e-9-perturbed identity and a nonzero row	PASS
Gc-total-2p1d	S5 fluid rows == Nabla_mu (T_m + T_Q)^{mu nu}, sqrt(-g) route (symbolic)	PASS
Gc-number-2p1d	charm row == Nabla.N, sqrt(-g) route (symbolic)	PASS
Gc-dM-2p1d	the second moment is passive: dM columns of the fluid + first-moment rows vanish identically, and their sources carry no dM	PASS
Gc-off-2p1d	charm off (n, nu, dM -> 0): S5's medium block and medium sources = = S2 exactly; the charm rows' medium columns and sources vanish, so charm = 0 is an invariant subspace on which S5 IS S2	PASS
R0-jl2wl	Julia -> WL conversion of the shipped matrices (0)	PASS
R-S2-1p1d-shipped	S2 1+1D generic EOS, dpp = dPP = 0 == SHIPPED one_d_viscous_ matrix (1d_viscous_cilindrical.jl), entry by entry (55 entries, symbolic)	PASS
R-S2-2p1d-generator	S2 2+1D, dpp = dPP = 0 == two_d_viscous_matrix_derived (2d_ viscous.jl, from derive_2p1d_viscous.wls), entry by entry (154 entries, symbolic)	PASS
R-S2-1p1d-MIS	S2 1+1D with dpp = 4/3 tauS, dPP = 0 == medium block of one_d_ viscous_matrix_fugacity (:MIS, read from the .jl), entry by entry (55 entries, symbolic)	PASS
R-S2-1p1d-MIS-ctl	control: dpp = 0 does NOT reproduce the :MIS medium block (the two 1-D media differ by delta_pipi)	PASS
R-S4S3-1p1d	S4 rows 1-2 == S3 (so the first-moment block of S4/S5 is the consistent closure)	PASS
X-2to1-1	S5 2+1D axisymmetric on the x axis: tau-momentum (symbolic)	PASS
X-2to1-2	S5 2+1D axisymmetric on the x axis: x-momentum == r-momentum (symbolic)	PASS
X-2to1-3	S5 2+1D axisymmetric on the x axis: y-momentum == 0 (symbolic)	PASS
X-2to1-4	S5 2+1D axisymmetric on the x axis: E^{yy} == r^2 E^{phiphi} (symbolic)	PASS
X-2to1-5	S5 2+1D axisymmetric on the x axis: E^{etaeta} (symbolic)	PASS
X-2to1-6	S5 2+1D axisymmetric on the x axis: E^{xy} == 0 (symbolic)	PASS
X-2to1-7	S5 2+1D axisymmetric on the x axis: bulk (symbolic)	PASS

id	what it tests	
X-2to1-8	S5 2+1D axisymmetric on the x axis: charm number (symbolic)	PASS
X-2to1-9	S5 2+1D axisymmetric on the x axis: nu^x row == nu^r row (symbolic)	PASS
X-2to1-10	S5 2+1D axisymmetric on the x axis: nu^y row == 0 (symbolic)	PASS
X-2to1-11	S5 2+1D axisymmetric on the x axis: piQ^{yy} row == (phi phi) row (symbolic)	PASS
X-2to1-12	S5 2+1D axisymmetric on the x axis: piQ^eta_eta row == (eta eta) row (symbolic)	PASS
X-2to1-13	S5 2+1D axisymmetric on the x axis: piQ^{xy} row == 0 (symbolic)	PASS
X-2to1-14	S5 2+1D axisymmetric on the x axis: trace row (symbolic)	PASS
X-ctrl	control: the same test REJECTS a wrong pairing (2+1D tau row vs 1+1D r row)	PASS
X-vort-axis	the corotation coupling W vanishes identically on an axisymmetric state (so G-chart is blind to wv, as documented)	PASS
X-1to0-1	S5 1+1D at u^r = 0, no r-dependence: energy (symbolic)	PASS
X-1to0-2	S5 1+1D at u^r = 0, no r-dependence: r-momentum == 0 (symbolic)	PASS
X-1to0-3	S5 1+1D at u^r = 0, no r-dependence: E^{phiphi} == -(tau^2/2r^2) E^{etaeta} (symbolic)	PASS
X-1to0-4	S5 1+1D at u^r = 0, no r-dependence: E^{etaeta} (symbolic)	PASS
X-1to0-5	S5 1+1D at u^r = 0, no r-dependence: bulk (symbolic)	PASS
X-1to0-6	S5 1+1D at u^r = 0, no r-dependence: charm number (symbolic)	PASS
X-1to0-7	S5 1+1D at u^r = 0, no r-dependence: nu^r row == 0 (symbolic)	PASS
X-1to0-8	S5 1+1D at u^r = 0, no r-dependence: (phi phi) row == -1/2 (eta eta) row (symbolic)	PASS
X-1to0-9	S5 1+1D at u^r = 0, no r-dependence: (eta eta) row (symbolic)	PASS
X-1to0-10	S5 1+1D at u^r = 0, no r-dependence: trace row (symbolic)	PASS
B-ideal-conf	conformal ideal Bjorken: dT/dtau == -T/(3 tau)	PASS
B-ideal-gen	generic-EOS ideal Bjorken: dT/dtau == -P'/(P" tau) == -c_s^2 T/tau, c_s^2 = P'/(T P")	PASS
B-IS-gen	generic IS Bjorken: T' = -(T P' + Pi + pi)/(tau T P"), tauS pi' = -pi - 4 visc/(3 tau) - dpp pi/tau, tauB Pi' = -Pi - zeta/tau - dPP Pi/tau	PASS
B-IS-conf	conformal IS Bjorken: dT/dtau = -T/(3tau) - pi/(12 a T^3 tau), tauS dpi/dtau = -pi - 4 visc/(3 tau) - dpp pi/tau [dpp = 4/3 tauS: the DNMR delta_pipi form; no tau_pipi, lambda_1]	PASS
B-NS-conf	first-order (NS) limit pi = -4 visc/(3 tau): dT/dtau = -T/(3tau) + (4/9)(eta/s)/tau^2, s = 4 a T^3	PASS
B-twoway-energy	S5 Bjorken energy row: d_tau(e + e_Q) + (e+P+Pi+pi + n h)/tau	PASS
B-charm-number	Bjorken charm number: (1/tau) d_tau (n tau) = 0	PASS
T-rest-1p1d	nu = 0, dM = 0, grad alpha = 0, medium at rest: transverse momentum exchange Delta.Nabla.T_Q == 0 and the nu row is satisfied	PASS
T-drag-1p1d	drag: on shell, Nabla.T_Q^i = -(T/Ds) nu^i (drag on charm) and the fluid row then gives Nabla.T_m^i = +(T/Ds) nu^i (Newton III), exact	PASS
T-split-1p1d	fluid momentum rows == Nabla.T_m + Nabla.T_Q identically	PASS
T-rest-2p1d	nu = 0, dM = 0, grad alpha = 0, medium at rest: transverse momentum exchange Delta.Nabla.T_Q == 0 and the nu row is satisfied	PASS
T-drag-2p1d	drag: on shell, Nabla.T_Q^i = -(T/Ds) nu^i (drag on charm) and the fluid row then gives Nabla.T_m^i = +(T/Ds) nu^i (Newton III), exact	PASS
T-split-2p1d	fluid momentum rows == Nabla.T_m + Nabla.T_Q identically	PASS
E-same-one_d_HQ_BG_consistent_matrix	simplified exported entries == derived entries (symbolic)	PASS
E-free-one_d_HQ_BG_consistent_matrix	every free symbol is an argument or a field	PASS
E-same-one_d_HQ_BG_m2_matrix	simplified exported entries == derived entries (symbolic)	PASS
E-free-one_d_HQ_BG_m2_matrix	every free symbol is an argument or a field	PASS
E-same-one_d_two_way_matrix	simplified exported entries == derived entries (symbolic)	PASS
E-free-one_d_two_way_matrix	every free symbol is an argument or a field	PASS
E-same-one_d_two_way_matrix_conformal	simplified exported entries == derived entries (symbolic)	PASS
E-free-one_d_two_way_matrix_conformal	every free symbol is an argument or a field	PASS

id	what it tests	
E-same-two_d_HQ_BG_consistent_matrix	simplified exported entries == derived entries (symbolic)	PASS
E-free-two_d_HQ_BG_consistent_matrix	every free symbol is an argument or a field	PASS
E-same-two_d_HQ_BG_m2_matrix	simplified exported entries == derived entries (exact at 3 random rational points)	PASS
E-free-two_d_HQ_BG_m2_matrix	every free symbol is an argument or a field	PASS
E-same-two_d_two_way_matrix	simplified exported entries == derived entries (exact at 3 random rational points)	PASS
E-free-two_d_two_way_matrix	every free symbol is an argument or a field	PASS
E-same-two_d_two_way_matrix_conformal	simplified exported entries == derived entries (exact at 3 random rational points)	PASS
E-free-two_d_two_way_matrix_conformal	every free symbol is an argument or a field	PASS
E-jl	the generated Julia (8 functions) parses, runs, and reproduces its 24-state CSVs: max rel err 1.4298582449629782e-15	PASS

102 passed, 0 failed (102 checks evaluated).

Not a check: the characteristic speeds of S5 are computed and reported (`two_way_char_speeds.tsv` in the scratch directory).

Re-run with `wolframscript -file deriv/derive_two_way.wls`.